Journal of Asian Earth Sciences 185 (2019) 104046

Contents lists available at ScienceDirect

Journal of Asian Earth Sciences

journal homepage: www.elsevier.com/locate/jseaes

Petrogenesis and tectonic setting of Late Paleozoic to Late Mesozoic igneous rocks in Cambodia

Rathborith Cheng[a,b], Etsuo Uchida[a,*], Masato Katayose[a], Kosei Yarimizu[a], Ki-Cheol Shin[c], Sitha Kong[b], Takanori Nakano[a]

[a] Department of Resources and Environmental Engineering, School of Creative Science and Engineering, Waseda University, Ohkubo 3-4-1, Shinjuku, Tokyo 169-8555, Japan
[b] Department of Geology, Ministry of Mines and Energy, #79-80, St. 51, Sankat Phsar Thmey III, Khan Daun Penh, Phnom Penh 12210, Cambodia
[c] Research Institute for Humanity and Nature, Kamigamo 457-4, Kita-ku, Kyoto 603-8047, Japan

ARTICLE INFO

Keywords:
Igneous rock
Cambodia
Tectonic setting
Chemical composition
Magnetic susceptibility
Sr-Nd isotope ratio

ABSTRACT

Emplacement of igneous rocks from the Carboniferous to the Middle Jurassic of Cambodia was controlled by the Indosinian Orogeny, which was related to the sequence of suturing and collision between the continental fragments from Gondwana. To constrain magma genesis, tectonic setting, and relationship with igneous rocks in neighboring countries, we determined the magnetic susceptibility, chemical compositions, Rb-Sr isochron ages and Sr-Nd isotope ratios for igneous rock samples from Cambodia. Igneous rocks in Cambodia can be divided by the inferred Mae Ping fault into magnetite series in the NE region and ilmenite series in the SW region. The igneous rocks in the NE region showed I-type or A-type and metaluminous signatures, and strong involvement of mantle source materials. They are subdivided into older and younger adakitic rocks and non-adakitic rocks. The igneous rocks in the SW region showed non-adakitic and slightly metaluminous to peraluminous signatures. These have significant involvement of source materials with relatively uniform Nd isotope ratios but variable Sr isotope ratios, which may have originated from the lower to middle continental crust. The Late Triassic to Middle Jurassic igneous rocks were formed by the subduction of the Paleo-Tethys Ocean crust associated with the collision of the Sibumasu Terrane with the Indochina Terrane (the Indosinian Orogeny II). The younger granitic rocks showed A-type signature and are regarded as post-orogenic, within the plate or rifting zone that intruded in association with the subduction of the Paleo-Pacific Ocean crust beneath Indochina during the Cretaceous to the Paleogene.

1. Introduction

Cambodia is situated on the Indochina Terrane in Southeast Asia which rifted northward from the northwestern part of the super-continent Gondwana during the Middle Devonian (Metcalfe, 2011, 2013). Cambodia borders Thailand to the west, Laos to the north, Vietnam to the southeast, and the Gulf of Thailand to the southwest (Fig. 1). Many igneous rocks are found in these countries, and their formation was strongly controlled by tectonic activity, namely the Indosinian Orogeny (Workman, 1977; Fontaine and Workman, 1978; ESCAP V.10, 1993; Carter et al., 2001). This orogeny is related to the collision of the fragments from Gondwana, the opening and closing of the Tethys Ocean, and oceanic plate subduction.

To reconstruct the geological history of Indochina, many studies have been conducted to examine magma types, tectonic setting and related mineralization of igneous rocks in Indochina (Thuy et al., 2004; Sanematsu et al., 2011; Metcalfe 2011, 2013, Morley, 2012; Kong et al., 2012; Morey et al., 2013; Kamvong et al., 2014; Zaw et al., 2014; Faure et al., 2018; Rossignol et al., 2018). However, there are few studies on igneous rocks in Cambodia. In this study we present a review of geological and petrological studies of Indochina and discuss regional characteristics of whole rock and mineral chemistries, Rb-Sr isochron ages and Sr and Nd isotopic geochemistry of igneous rocks distributed throughout Cambodia.

2. Geological setting and igneous rocks in Indochina

2.1. Geological history

Southeast Asia is a collage of allochthonous continental blocks and fragments (terranes) connected by suture zones or by geological discontinuities such as major strike-slip faults (Fig. 2) (e.g., Sone and

* Corresponding author.
E-mail address: weuchida@waseda.jp (E. Uchida).

https://doi.org/10.1016/j.jseaes.2019.104046
Received 15 December 2018; Received in revised form 25 September 2019; Accepted 26 September 2019
Available online 27 September 2019
1367-9120/ © 2019 Elsevier Ltd. All rights reserved.

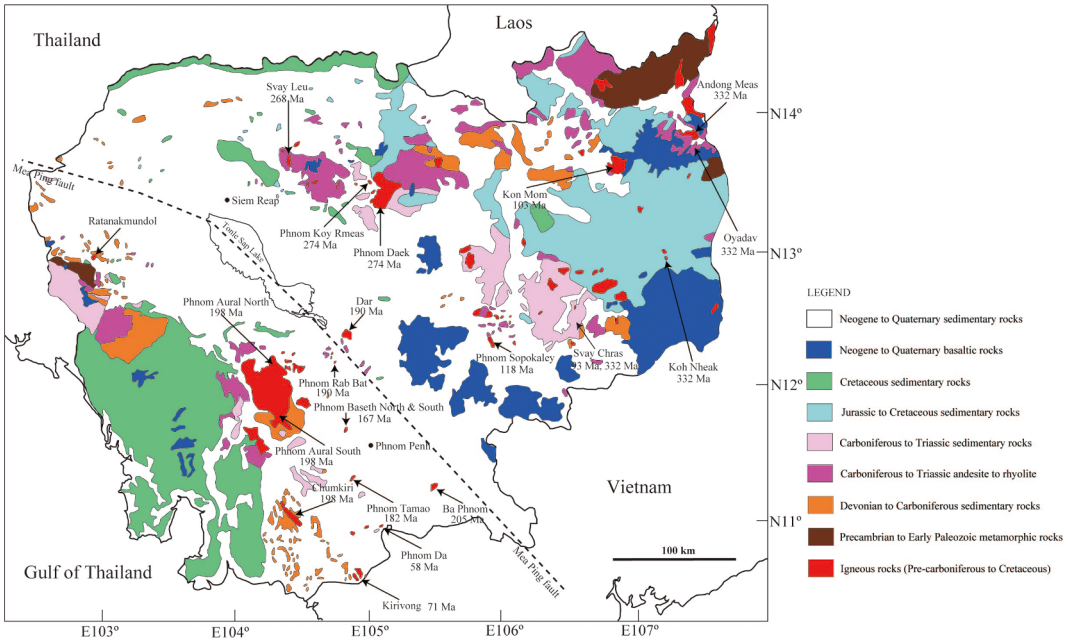


Fig. 1. Geological map of Cambodia based on Tien et al. (1990) and ESCAP (1993), showing the studied igneous rocks. The Rb-Sr isochron ages of the igneous rocks obtained in this study are also shown.

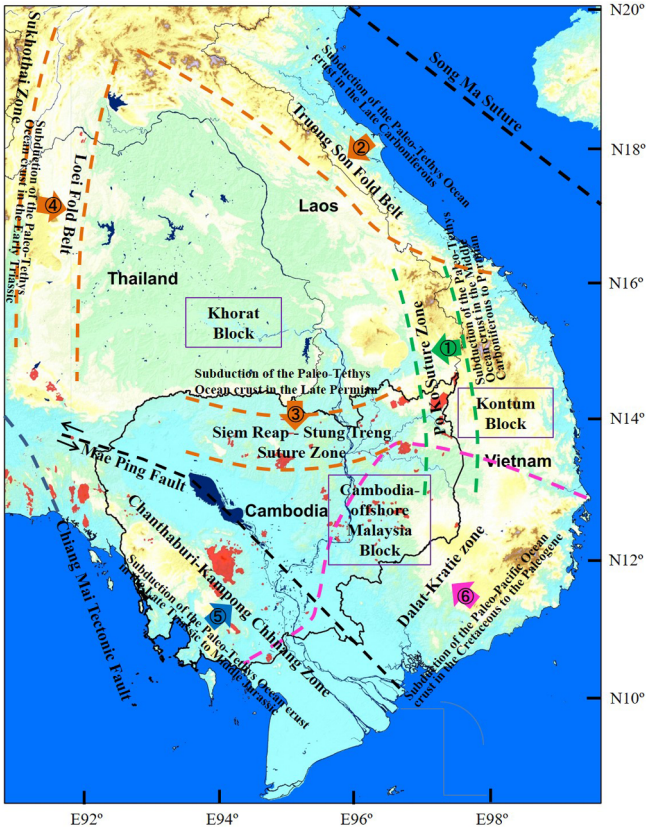


Fig. 2. Tectonic settings map in SE Asia deduced from the igneous rocks. The circled numbers in the figure show the order of subduction of oceanic crusts.

Metcalfe, 2003; Metcalfe 2013; Zaw et al., 2014). The Gondwana-derived continental blocks now located in SE Asia evolved during the mid-Paleozoic to the Cenozoic; this involved three phases of rifting and separation. There were three Tethys Oceans: the Paleo-Tethys, Meso-Tethys, and Ceno-Tethys Oceans which opened behind the separating blocks and subsequently closed. The Indochina Terrane and the South and North China Terranes rifted from Gondwana to open the Paleo-Tethys Ocean in the Middle Devonian. The closure of the Paleo-Tethys Ocean by northward subduction beneath the Indochina Terrane in the Early Permian separated the Sukhothai arc from the Indochina Terrane by opening the Nan-Uttaradit Basin. During this event, volcanic-arc-related I-type igneous rocks were emplaced. This back-arc ocean was narrow and short-lived (from the Early Permian to the Early Triassic) because of new rifting of the Sibumasu Terrane from Gondwana in the Early Permian which opened the Meso-Tethys Ocean (Metcalfe, 2011, 2013). The Ceno-Tethys Ocean opened by rifting and separation of the West Myanmar Terrane in the Late Triassic to Late Jurassic, which accreted with the Sibumasu Terrane in the late Cretaceous. Finally, the Indian subcontinent rifted from Gondwana in the Cenozoic (Metcalfe, 2011, 2013) and collided with the Eurasia continent from 50 Ma to 40 Ma.

The Indochina Terrane is the largest tectonic unit in SE Asia; its tectonics were affected by the Truong Son, Loei and Dalat-Kratie fold belts (Fig. 2). In SW Cambodia, the Chanthaburi-Kampong Chhnang zone corresponds to the Loei fold belt (e.g., Sone and Metcalfe, 2003; Nualkhao et al., 2018). The Indochina Terrane was separated into several small blocks by the opening of several branches of the Paleo-Tethys Ocean. The north-striking Po Ko Suture zone, situated in the western part of the Kontum block in central Vietnam, divides the Indochina Terrane into western and eastern parts. West-dipping subduction and collision of the Kontum block (eastern Indochina) occurred along the Po Ko Suture zone beneath the eastern margin of the Khorat block (western Indochina) during the Late Carboniferous to the Permian (the Early Indosinian Orogeny) (Lepvrier et al., 2004, 2008). The Indochina Terrane consists of a series of continental blocks which were amalgamated by closure along older rifts during the Early Triassic (the Indosinian Orogeny I). The back-arc basin between the Indochina Terrane and the Sukhothai arc closed simultaneously with or slightly after the collision between the South China–Cathaysia Terrane and the Indochina Terrane (Morley et al., 2013). The Sibumasu Terrane collided with the Sukhothai-Indochina Terrane in the Late Triassic (the Indosinian Orogeny II).

The eastern margin of the Indochina Terrane in southern Vietnam (the Dalat-Kratie zone) is interpreted as resulting from the subduction of the Paleo-Pacific Ocean crust beneath Indochina during the Cretaceous (Thuy et al., 2004; Morley, 2012). These Cretaceous (125–75 Ma) igneous rocks are widely distributed from southeastern

Cambodia to southern Vietnam (Kamvong et al., 2014; Zaw et al., 2014).

2.2. Igneous rock studies

Many studies have analyzed and discussed formation ages and magma types of granitic rocks in Indochina. Kamvong et al. (2014) conducted U-Pb zircon dating for diorite and monzodiorite in the Troung Son and Loei fold belts in Thailand and Laos. Their research revealed that the Phu Kham adakitic rocks in the Truong Son belt were emplaced during the Late Carboniferous (ca. 306–304 Ma), whereas the Puthep 1 (PUT 1) adakitic rocks in the Loei fold belt were formed during the Middle Triassic (ca. 244–241 Ma). Tectonically, the Phu Kham and PUT 1 adakitic rocks were most likely formed during initiation of subduction of the Ailaoshan-Song Ma and main Paleo-Tethys Oceans, respectively. There is a close temporal relationship between the adakitic rocks and Cu-Au mineralization in the Phu Kham and PUT 1 deposits (Kamvong et al., 2014). Sanematsu et al. (2011) conducted $^{40}Ar/^{39}Ar$ age determination of granitic rocks in the Truong Son fold belt and Kontum block (massif) in Laos. They also found I-type and magnetite series granitic rocks in the Kontum block in southeasternmost Laos.

Thuy et al. (2004) reported that the Cretaceous granitic rocks in the Dalat-Kratie zone in southern Vietnam were of sub-alkaline affinity belonging to the high-K calc-alkaline-series and displayed features of typical I-type granitic rocks. They found that the Dinhquan suite consists of hornblende-biotite granodiorite, diorite, and minor granite emplaced at 112 Ma to100 Ma while the Deoca suite comprises pink porphyritic hornblende-biotite-bearing granodiorite, monzogranite, and diorite with ages from 92 Ma to 88 Ma. The Cana suite is biotite-bearing granite, which is poor in hornblende with strong negative Eu anomalies and ages around 96–93 Ma. Based on these data, they concluded that the granitic rocks in the Dalat-Kratie zone formed as a result of the subduction of the Paleo-Pacific Ocean crust beneath Indochina during the Cretaceous (Thuy et al., 2004; Morley, 2012). Shellnutt et al. (2013) concluded that the Cana suite (the Ankroet suite) in the Dalat-Kratie zone is post-orogenic/within-plate granitic rock (A-type granitic rock).

Kong et al. (2012) studied igneous rocks in Cambodia, conducting magnetic susceptibility measurements and whole rock geochemistry analysis of igneous rocks from eight regions, mostly in SW Cambodia. They reported that the granitic rocks could be divided into magnetite series in the north and ilmenite series in the south, although the boundary between the two rock types is unclear. There are few studies on the mineral chemistry and isotopic geochemistry of igneous rocks in Cambodia, and accordingly their formation environments are poorly understood.

3. Samples and analysis

3.1. Distribution, magnetic susceptibility, and samples of igneous rocks

The topography of Cambodia includes mountainous regions such as the Cardamom Highlands in the western region, Khorat Plateau in the northern region, and Kontum massif in the eastern region. Plains are widely distributed in central Cambodia; they were formed by the Mekong River in the east, flowing north to south, and the Tonle Sap River in the northwest flowing out of Lake Tonle Sap. The two rivers join in Phnom Penh, Cambodia's capital city. The Mae Ping fault runs from Thailand in the northwest through Lake Tonle Sap in Cambodia to southern Vietnam (Morley et al., 2013).

The distribution of igneous rocks in Cambodia is shown in Fig. 1. In this study, we collected igneous rocks from two regions – NE Cambodia and SW Cambodia, with the Mae Ping fault separating the two regions. The sampling sites in NE Cambodia were Phnom Daek, Phnom Koy Rmeas, Svay Leu, Svay Chras, Phnom Soporkaley, Kon Mom, Koh Nheak, Andong Meas and Oyadav, and those in SW Cambodia were Ba Phnom, Phnom Da, Kirivong (Tonleap and Phnom Den), Phnom Tamao, Phnom Baseth North and South, Chumkiri, Phnom Aural North and South, Dar, Phnom Rab Bat, and Ratanakmundol. These locations were in quarries, mining areas, national parks, pagodas, villages, or outcrops along roads. Several igneous rock outcrops could not be accessed due to bad road conditions or landmines.

At each outcrop, we measured the magnetic susceptibility of the igneous rocks using an SM30 portable magnetic susceptibility meter (ZH Instruments, Brno, Czech Republic). The measurements were conducted on 10 flat and non-weathered surfaces at each outcrop. We collected 26 samples from 9 locations in NE Cambodia, and 34 samples from 12 locations in SW Cambodia.

3.2. Sample preparation and analysis

Fresh samples were taken from each outcrop for laboratory analyses. Thin sections were prepared from each sample for inspection under a polarizing microscope for a petrographic description. Chemical composition analysis of biotite was conducted using a scanning electron microscope, JEOL JSM-6360, equipped with an energy dispersive X-ray spectrometer INCA-Energy (SEM-EDX) (Oxford Instruments, UK). The polished thin sections were coated with carbon before analyses.

The rock samples were pulverized for 1 min using a tungsten carbide rod mill. About 5 g of the rock powder were sent to Activation Laboratories Ltd. (Ancaster, Canada) for whole rock chemical composition analysis using their "4Litho" litho-geochemistry package. For these analyses, the rock powders were fused using lithium metaborate/tetraborate and digested in dilute nitric acid. The analyses covered 55 elements using an inductively coupled plasma atomic emission spectrometer (ICP-AES) and inductively coupled plasma mass spectrometer (ICP-MS).

We determined the isotopic ratios of Sr and Nd for representative rock samples at the Research Institute for Humanity and Nature (RIHN) in Kyoto. Around 0.1 g of pulverized samples were dissolved using mixing solutions of nitric acid (HNO_3), hydrofluoric acid (HF), and perchloric acid ($HClO_4$) in a digestion bomb with a liner made of poly tetra fluoro ethylene (PTFE) at 160 °C. The Sr and Nd were separated using appropriate resin loaded columns. After separation, the isotopic composition of each element was measured by a plasma-source mass spectrometer (Multicollector ICP-MS NEPTUNE, Thermo Fisher Scientific, Massachusetts, USA). For more details of the Sr and Nd separation and standard samples, refer to Shin et al. (2009).

4. Results and discussion

4.1. Magnetic susceptibility

The magnetic susceptibility of igneous rock depends mainly on the presence of magnetite and its abundance. Granitic rocks can be classified by magnetic susceptibility into the magnetite series with values higher than 3.0×10^{-3} SI units, and the ilmenite series with lower values (Ishihara, 1977). The magnetic susceptibility of the igneous rocks in Cambodia ranged widely from 0.01×10^{-3} to 100×10^{-3} SI units (Fig. 3). In the NE region through to the Mae Ping fault, the magnetic susceptibility was higher than 3.0×10^{-3} SI units, and in the SW region it was lower, indicating that the igneous rocks in the NE region belong to the magnetite series while the ones in the SW region fit within the ilmenite series. Igneous rocks such as the granitic rocks of Ba Phnom, Dar, Phnom Rab Bat, and Phnom Aural North situated near the inferred Mae Ping fault have magnetic susceptibilities near the boundary between the two series. Thus, igneous rocks in Cambodia can be divided along this fault.

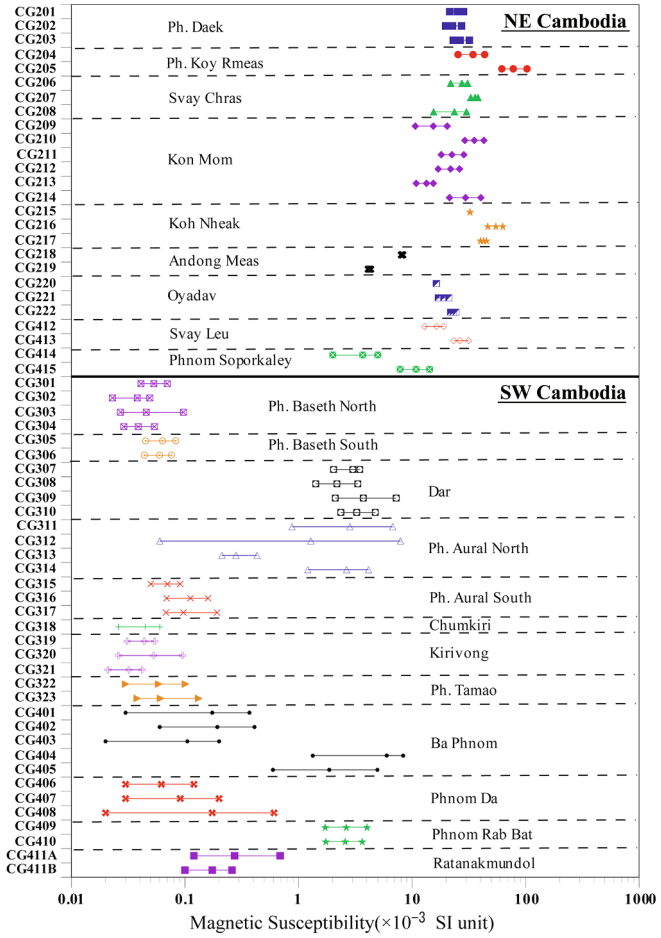


Fig. 3. Magnetic susceptibilities of the igneous rocks in Cambodia.

4.2. Geochemical characteristics

4.2.1. Classification of igneous rocks

The constituent minerals and whole rock chemical compositions of the igneous rock samples are summarized in Tables 1 and 2, respectively. Co and W were deleted from Table 2 because of contamination from the rod mill. The igneous rocks in Cambodia plotted on the SiO_2 vs. $Na_2O + K_2O$ diagram (Fig. 4), according to classification by Cox et al. (1979) and Wilson (1989), show that they are composed of both the alkalic series and sub-alkalic series, but their SiO_2 concentrations differ greatly between the NE region and the SW region. The igneous rocks in the NE region range from gabbro to granite, indicating wide variations in SiO_2 concentrations. Several igneous rocks in the NE region also have high concentrations with respect to $Na_2O + K_2O$ contents. They are composed of syenite (Kon Mom), syenite-diorite (Kon Mom, and Svay Chras), and gabbro (Koh Nheak) in the alkaline series. The igneous rocks in the SW region, however, are composed of granite and small amounts of granodiorite. Several igneous rocks in the SW region are highly differentiated and have high SiO_2 concentrations (over 75 wt%).

This variation in the chemical composition of the igneous rocks agrees well with their mineral composition. Except for the Oyadav site, the granite and granodiorite in the NE region are rich in plagioclase (Pl) and poor in potassic feldspar (Kf) while those in the SW region are high in Kf and low in Pl. The igneous rocks in the NE region are rich in hornblende (Hb) and do not contain muscovite (Ms), whereas the Phnom Baseth (North and South) and Kirivong samples in the SW region contain Ms. Opaque minerals, mainly magnetite, are abundant in the igneous rocks in the NE region but are scarce in the ones in the SW region. The syenite-diorites, which are present in the NE region, are rich in Pl, Kf and the mafic minerals; clinopyroxene (Cpx), hornblende (Hb), and biotite (Bi). Especially in Kon Mon, the syenite and syenite-diorite contain high levels of Kf. The gabbro of Phnom Koy Rmeas is of the sub-alkaline series and has quartz (Q) and high Hb content, but that of Koh Nheak is of the alkaline series and contains a moderate amount of Hb and no Q.

4.2.2. Petrography

NE Cambodia: The igneous rock in Phnom Daek is granodiorite to granite with a medium grain size, showing an equigranular texture (Fig. 5a and b). It contains abundant Pl, moderate amounts of Q, Bi, and Hb, and small amounts of Kf. Bi, Pl and Hb are frequently altered. Phnom Daek igneous rock is accompanied by a skarn-type iron deposit.

The igneous rock in Phnom Koy Rmeas is gabbro with a coarse grain size and has abundant Pl and Hb, and small amounts of Q and Cpx.

The igneous rock in Svay Leu is granite to granodiorite with a medium grain size. The igneous rock consists mainly of abundant Pl, moderate amounts of Q, Bi and Hb, and minor Kf.

Svay Chras is actively quarried. The igneous rock in Svay Chras is syenite-diorite with a medium grain size and contains abundant Pl, moderate amounts of Cpx and Bi, and small amounts of Q, Kf, and Hb.

The igneous rock in Phnom Soporkaley is granodiorite with a medium grain size that has abundant Pl, moderate amounts of Cpx, Q, and Hb, and small amounts of Bi and Kf.

The igneous rock in Kon Mom is syenite-diorite with a medium to coarse grain size (Fig. 5c and d). It contains considerably abundant Kf; moderate amounts of Pl, Hb, Cpx, and Bi; and a small amount of Q.

The igneous rock in Koh Nheak is melanocratic gabbro with a medium grain size and consists mainly of abundant Pl, a moderate amount of Hb, and small amounts of Cpx and olivine (Ol) (Fig. 5e and f).

The igneous rock in Andong Meas is granodiorite with a medium grain size and contains abundant Q and Pl, a moderate amount of Bi, and a small amount of Kf.

The igneous rock in Oyadav is granodiorite with a medium grain size. It consists mainly of moderate amounts of Q, Pl, Kf, and Hb. Granophyric texture is observed.

Gold exploration is ongoing in the igneous rocks of Koh Nheak and Andong Meas, and Cu-Au exploration is occurring in the igneous rocks in Oyadav by Angkor Gold Corp.

SW Cambodia: The igneous rock in Ba Phnom is granite with a medium grain size, and consists mainly of abundant Kf, moderate amounts of Q, Pl, and Bi, and a small amount of Hb.

The igneous rock in Phnom Da is leucogranite with a medium grain size and consists mainly of abundant Q and Kf, a moderate amount of Pl, and a small amount of Bi.

There are two types of igneous rocks in Kirivong. One (Tonleap) is porphyritic granite with a medium grain size and contains Pl and Kf as phenocrysts (Fig. 6a and b). It consists mainly of abundant Q and Bi, with moderate amounts of Pl and Kf. Another igneous rock is located in Phnom Den in the eastern part of the Kirivong area. It is granodiorite with a fine to medium grain size. The granodiorite consists mainly of abundant Q, moderate amounts of Pl and Kf, and a small amount of Bi.

The igneous rock in Phnom Tamao is porphyritic granite and consists mainly of abundant Kf and Bi, and moderate amounts of Q and Pl.

The igneous rock in Phnom Baseth North is granite with a porphyritic texture (Fig. 6c and d). The Phnom Baseth North granite consists mainly of abundant Q and Kf, and moderate amounts of Pl and Bi. Mineralization of Mo and Cu is confirmed in Phnom Baseth North granite (Kong et al., 2011). The Phnom Baseth South granite contains abundant Q, Pl, and Bi, and a moderate amount of Kf.

The igneous rock in Chumkiri is strongly altered granite and was not

Table 1
Constituent minerals of igneous rock samples collected in this study.

	Sample No.	Rock type	Q	Pl	Kf	Bi	Hb	Zr	Ap	Mu	Ti	Op	Ep	Ru	Cpx	Tour	Alla	Cal	Remarks
Phnom Daek	CG201	Granodiorite	◎	◎	△	○	○		–		–	–							Bi is parly altered. Pl is altered.
	CG202	Granodiorite	○	◎	△	○	○	–	–		–	–							Bi is parly altered.
	CG203	Granite	○	◎	△	○	○					–							Bi and Pl are altered. Hb is partly altered
Phnom Koy Rmeas	CG204	Gabbro	△	◎			◎					△							
	CG205	Gabbro	△	◎			◎					△			△				
Svay Chras	CG206	Syenite-diorite	△	◎	–	○			–			–			○				
	CG207A	Syenite-diorite	△	◎	△	○	△		–			–	–		○				
	CG207B	Syenite-diorite	△	◎	△	○	△		–			–			○				
	CG208	Syenite-diorite	△	◎	△	△	△		–			–			○				Bi is altered.
Kon Mom	CG209A	Syenite	–	△	◎	△	○					–						–	
	CG209B	Syenite	△	△	◎	△	○	–				–						–	
	CG210	Syenite-diorite		○	◎	△			–			–			○				Bi is almost altered.
	CG211	Syenite-diorite		○	◎	○		–	–			–			○				
	CG212	Syenite-diorite	△	○	◎	○	△	–	–			–			○				
	CG213	Syenite-diorite	△	◎	○	○	○		–			–			△				
	CG214	Syenite-diorite	△	◎	◎	○	△		–			–			○	–			
Koh Nheak	CG215	Gabbro		◎			○					△			△				Pl is strongly altered.
	CG216	Gabbro		◎			○					△			△				Pl is strongly altered.
	CG217	Gabbro		◎		△	○					△			△				Olivine occurs
Andong Meas	CG218	Granodiorite	◎	◎	△	○			–			–							
	CG219	Granodiorite	◎	◎		○			–			–	–						
Oyadav	CG220	Granodiorite	○	○	○		○				–	–	–						
	CG221	Granodiorite	○	○	○		○				–	–							
	CG222	Granodiorite	○	○	○		○				–	–	–						Ep occurs as alteration mineral.
Svay Leu	CG412	Granite	◎	◎		○	–					–							
	CG413	Granodiorite	○	◎		○	○				–	–	–						
Phnom Soporkaley	CG414	Granodiorite	○	◎		△	△	–	–			–			○				
	CG415	Granodiorite	○	◎	○		○					–	–		△	–			Strongly altered, Granophyric structure
Phnom Baseth North	CG301	Granite	◎	○	◎	○		–		△		–							Bi is partly alterd.
	CG302	Granite	◎	○	◎	○		–		–		–				–			
	CG303	Granite	◎	○	◎	○		–	–	–		–				–			Bi is partly alterd. Mu occurs as secondary mineral.
	CG304	Granite	◎	○	◎	○		–		–								–	Bi is partly alterd. Mu occurs as secondary mineral.
Phnom Baseth South	CG305	Granite	◎	◎	○	◎		–		–		–							Mu occurs as secondary mineral.
	CG306	Granite	◎	○	○	◎		–	–	–	–	–							Mu occurs as secondary mineral.
Dar	CG307	Granite	○	○	◎	○	△	–				–			–	–			Bi is partly alterd.
	CG308	Granite	○	○	◎	○		–				–				–			Bi is partly alterd.
	CG309	Granite	○	○	◎	○	△	–	–			–				–			Bi is partly alterd.
	CG310	Granite	○	○	◎	○						–	–			–			All Bi is altered. Kf is altered.
Phnom Aural North	CG311	Granite	◎	○	○	○		–	–			–							Bi is partly alterd.
	CG312	Granite	◎	○	○	△												–	Bi and Cal are partly alterd.
	CG314	Granite	◎	○	◎	○			–			–							Bi is partly altered.
Phnom Aural South	CG315	Granite	◎	○	◎	◎	△	–	–										
	CG316	Granite	○	○	◎	◎	○	–	–				–			–			
	CG317	Granite	◎	○	◎	○	○	–	–			–							
Chum Kiri	CG318	Granite	◎	◎	○	△						–						–	Cal is altered. All Bi is altered.
Kirivong	CG319	Granite	◎	○	○	◎		–				–							
	CG320	Granite	◎	○	○	○		–		△		–							
	CG321	Granodiorite	◎	○	○	△				△		–							Bi is partly alterd.
Phnom Tamao	CG322	Granite	○	○	◎	◎		–	–			–							
	CG323	Granite	○	○	○	◎		–				–							
Ba Phnom	CG401	Granite	○	○	○	○	○					–							
	CG402	Granite	○	○	◎	○	△	–	–			–							
	CG403	Granite	○	○	◎	○	△	–				–							
	CG404	Granite	◎	△	◎	△		–				–				–			Magnetite-series
	CG405	Granite	◎		○		△					–	–						Ep occurs as alteration mineral, Magnetite series
Phnom Da	CG406	Granite	◎	○	◎	○													
	CG407	Granite	◎	◎	◎	△		–	–			–			–				
	CG408	Granite	◎	○	◎	○		–				–							

(*continued on next page*)

Table 1 (*continued*)

	Sample No.	Rock type	Q	Pl	Kf	Bi	Hb	Zr	Ap	Mu	Ti	Op	Ep	Ru	Cpx	Tour	Alla	Cal	Remarks
Phnom Rab Bat	CG409	Granite	○	○	◎	○	–	–	–			–				–			Strongly altered
	CG410	Granite	○	○	◎	○	–	–				–							Cal and Bi are almost altered
Ratanak-mundol	CG411A	Granite	◎	◎														△	Brecciated rocks
	CG411B	Granite	◎	◎								–						△	

◎: > 30 vol%, ○: 30–10 vol%, △: 10–2 vol%, and –: < 2 vol%.
Abbreviations: Q, Quartz; Pl, Plagioclase; Kf, Potassium feldspar; Bi, Biotite; Hb, Hornblende; Zr, Zircon; Ap, Apatite; Mu, Muscovite; Ti, Titanite; Op, Opaque minerals; Ep, Epidote; Ru, Rutile; Cpx, Clinopyroxene; Tour, Tourmaline; Alla, Allanite; Cal, Calcite.

used for the chemical analysis of minerals. Q, Pl, and Kf are mainly observed.

The Phnom Aural North granite has abundant Q, and moderate amounts of Kf, Pl, and Bi. The Phnom Aural South granite, which contains abundant Q, Kf, and Bi, and moderate amounts of Pl and Hb.

The igneous rocks in Dar (Fig. 6e and f) and Phnom Rab Bat are pink colored granites with medium grain size. They have similar mineral compositions, and consist mainly of abundant Kf, moderate amounts of Q, Pl, and Bi, a small amount of Hb, and minor Cpx.

The igneous rock in Ratanakmundol is strongly sheared and altered granite. It has a medium grain size. Abundant Q and Pl were observed.

4.2.3. I-, S-, and A-type classification

Beside the magnetite series and ilmenite series, other classifications have been proposed for granitic rocks. Chappell and White (1974, 2001) proposed that granitic rocks can be classified into I-type and S-type on the $Al_2O_3/(Na_2O + K_2O)$ (A/NK; alkalinity) vs. $Al_2O_3/(CaO + Na_2O + K_2O)$ (A/CNK; alumina saturation index: ASI) diagram (Fig. 7) and the Na_2O vs. K_2O diagram (Fig. 8). According to this classification, I-type granitic rocks were formed by partial melting of igneous protolith. They are metaluminous or slightly peraluminous with ASI < 1.1, and Na_2O > 3.2% at 5% K_2O and Na_2O > 2.2% at 2% K_2O. In contrast, S-type granitic rocks were formed by melting of meta-sediment. They are peraluminous with ASI > 1.1.

Another group of granitic rocks is the A-type group, proposed by Loiselle and Wones (1979) for calc-alkaline to alkaline rocks that occur in rift zones in stable continental blocks. They are enriched in incompatible trace elements such as large ion lithophile elements (LILEs) and high field strength elements (HFSEs), but low in trace elements compatible with mafic silicates (Co, Sc, Cr, and Ni) and plagioclase (Ba, Sr, and Eu). Moreover, they are characterized by high SiO_2, $Na_2O + K_2O$, Fe/Mg, Ga/A1, Zr, Nb, Ga, Y, and Ce contents, and low CaO and Sr contents (Whalen et al., 1987; Bonin, 2007). A-type granitic rocks can be discriminated from I- and S-type granitic rocks using a Zr vs. 10,000 × Ga/Al diagram (Whalen et al., 1987) (Fig. 9). A-type granitic rocks are thought to have been formed under low H_2O pressures, extensional tectonic settings, and relatively high HF/H_2O conditions, and were dominantly non-crustal in origin and derived from strong differentiation of variously contaminated mantle-derived alkali basalts.

We describe the igneous rock types in the NE and SW regions based on their classification indexes.

NE Cambodia: The syenite-diorites of Svay Chras and Kon Mom are mostly classified as A-type in the Zr vs. 10,000 × Ga/Al diagram (Fig. 9), but they are not typical A-type due to the low FeO/MgO and a weak negative Eu anomaly. The rest of the igneous rocks in the NE region have lower A/CNK and higher A/NK than the ones in the SW region, and are classified as I-type in the alkalinity vs. ASI diagram (Fig. 7) and Na_2O vs. K_2O diagram (Fig. 8). The igneous rocks in the NE region are metaluminous except for those in Andong Meas and Svay Leu (one sample) which are peraluminous (Fig. 7).

SW Cambodia: The igneous rocks in the SW region have higher Ga/Al and more A-type characteristics than the ones in the NE region. For example, the granites in Kirivong except for Phnom Den (CG321), Phnom Da, Phnom Tamao, and Ba Phnom are A-type (Fig. 9). The total REY (REEs + Y) contents are high in A-type Kirivong, Ba Phnom, and Phnom Tamao granites: up to 1009 ppm, 1038 ppm, and 659 ppm, respectively. Most of the rest of the igneous rocks in the SW region are I-type in the alkalinity vs. ASI and Na_2O vs. K_2O diagrams, but some samples from Phnom Baseth North are classified as S-type or A-type (Figs. 7 and 8). Because the granites in Phnom Baseth South are similar in mineralogical and petrochemical compositions to those in Phnom Baseth North, we classified both granites as I-S-A-type. One sample of the Kirivong granitic rocks (CG321: Phnom Den granodiorite) and one of the Chumkiri granites were classified as I-type in the Na_2O vs. K_2O diagram but as S-type in the alkalinity vs. ASI diagram.

4.2.4. Tectonic settings by Pearce's classification

Pearce et al. (1984) suggested that the tectonic environment responsible for the formation of granitic rocks differs depending on trace elements. They divide the tectonic settings of granitic rocks into four groups based on the Rb vs. Y + Nb and Rb vs. Yb + Ta diagrams (Fig. 10): ocean ridge granites (ORG) (M-type), volcanic arc granites (VAG) (I-type & M-type), within-plate granites (WPG) (A-type), and *syn*-collision granites (*syn*-COLG) (S-type).

Using this classification index, most of the igneous rocks in the NE region fall within the VAG group (Fig. 10). One exception is the syenite-diorites in Kon Mom, which are classified as *syn*-COLG except for one sample of VAG. In contrast, the igneous rocks in the SW region tend to be more enriched in Rb than the ones in the NE region, due to the predominance of granites. Many granites are classified as *syn*-COLG, but some belong to WPG (Kirivong, Phnom Tamao, Ba Phnom, and Phnom Da) and VAG (Phnom Aural North and South, and Chumkiri). The Ratanakmundol granite could not be classified with this method because of the extremely low Rb content.

4.2.5. Adakitic rocks

According to Defant and Drummond (1990), adakitic rocks are derived from partial melting of young and hot ocean plates or ridges. They are of I-type and magnetite series, whose chemical compositions share trondhjemite-tonalite-dacite (TTD) compositions. High heat flow, rapid convection, and subduction of hot plates or ridges are essential tectonic elements required for partial melting of subducting ocean crust. Anatexis of Archean mid-ocean ridge basalt (MORB) under eclogite to garnet amphibolite conditions produces weakly peraluminous to metaluminous and high-Al TTD with low heavy rare earth elements (HREEs), Y, Nb, K/Rb, and Rb/Sr; and high La/Yb and Sr/Y contents. Such chemical features characterize adakitic rocks. Adakitic granitic rocks are frequently associated with Cu-Au mineralization (e.g., Kamvong et al., 2014). Fig. 11 shows a plot of igneous rocks in

Table 2
Whole rock chemical composition of igneous rock samples collected in this study.

Location	Phnom Deak			Ph. Koy Rmeas		Svay Chras				Kon Mom			
Sample No.	CG201	CG202	CG203	CG204	CG205	CG206	CG207A	CG207B	CG208	CG209A	CG209B	CG210	CG211
SiO_2(wt %)	65.53	63.82	71.67	52.28	47.57	53.93	52.28	53.15	54.97	63.56	65.93	60.46	57.4
Al_2O_3	15.05	15.63	12.67	17.31	17.78	16.24	13.29	16.07	18.81	17.52	17.04	17.86	17.12
Fe_2O_3(T)	4.07	5.31	4.3	9.55	13.76	8.28	12.1	9.3	6.45	3.09	2.19	4.73	6.53
MnO	0.082	0.082	0.132	0.166	0.2	0.115	0.191	0.139	0.092	0.216	0.107	0.203	0.12
MgO	2.27	2.64	1.07	6	5.58	3.53	6.48	4.7	2.36	0.51	0.29	1.33	3.54
CaO	4.66	5.08	2.62	10.63	11.65	6.41	7.32	7.57	7.74	1.29	0.79	3.74	4.9
Na_2O	3.82	3.67	3.91	2.65	2.01	3.91	3.28	3.86	4.16	5.27	5.06	4.89	4.24
K_2O	2.08	2.8	2.2	0.19	0.14	2.66	1.55	1.52	1.92	6.78	6.97	5.67	4.46
TiO_2	0.42	0.653	0.612	0.652	1.133	1.666	2.372	1.8	1.399	0.572	0.353	0.683	1.11
P_2O_5	0.16	0.17	0.12	0.08	0.21	0.6	0.85	0.71	0.58	0.11	0.1	0.3	0.37
LOI	1.14	0.89	0.71	0.89	0.73	1.16	0.39	0.5	0.37	0.69	0.78	0.68	0.97
Total	99.28	100.8	100	100.4	100.8	98.5	100.1	99.32	98.85	99.61	99.62	100.6	100.8
Sc(ppm)	10	13	14	30	37	18	23	20	12	4	4	8	13
Be	1	1	1	<1	<1	2	2	2	2	7	10	4	6
V	80	138	61	265	380	187	244	201	124	76	26	96	138
Ba	384	387	403	80	42	471	294	296	495	29	25	921	484
Sr	542	545	254	448	464	626	579	687	890	35	30	661	538
Y	13	20	40	17	12	23	29	25	18	42	45	23	24
Zr	119	193	235	43	26	287	186	154	242	247	363	437	394
Cr	20	50	<20	90	40	90	140	120	<20	<20	<20	<20	130
Ni	30	30	<20	50	30	90	90	80	<20	<20	<20	<20	60
Cu	20	40	40	40	190	60	90	110	60	<10	<10	30	90
Zn	40	40	90	70	80	100	130	120	70	120	90	120	80
Ga	16	17	13	16	19	21	19	25	22	23	20	19	20
Ge	1	1	2	1	1	1	1	1	<1	2	2	1	1
As	<5	<5	7	<5	<5	<5	<5	6	<5	<5	<5	<5	<5
Rb	63	63	40	2	<2	74	48	82	57	280	253	206	251
Nb	2	3	3	<1	<1	23	32	27	20	18	14	19	33
Mo	<2	<2	<2	<2	<2	2	3	3	<2	2	3	<2	6
Ag	<0.5	<0.5	<0.5	<0.5	<0.5	0.7	<0.5	0.9	<0.5	1.1	0.8	1	0.8
In	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2
Sn	<1	<1	1	<1	<1	2	2	3	1	3	2	1	2
Sb	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	1.2	<0.5	<0.5
Cs	2.3	1	1.1	<0.5	<0.5	3.2	2.2	6.4	3	3.5	5.4	6.2	13.8
La	20.6	27.9	12.7	5.2	3.5	36.5	44.5	43.2	33.1	85.1	70.4	31.2	44.3
Ce	41.7	59	32	13.8	9.1	77.1	95.8	93.8	67.8	177	129	62.9	91.2
Pr	4.78	7.05	4.56	2.09	1.4	9.13	11.5	11.1	8.18	19.9	14	7.67	10.2
Nd	17.9	26.9	21	10	7.1	36.5	46.7	43.6	33.5	73.2	50.4	30.7	38.7
Sm	3.5	5.4	5.8	2.8	2.2	7.5	9.6	9.3	6.5	14.1	9.5	6.1	7
Eu	0.93	1.29	1.31	0.9	0.93	1.88	1.85	2.06	2.16	1.69	1.24	1.89	1.58
Gd	2.9	4.4	6.3	3	2.6	6.7	8.3	7.5	5.4	10.2	8.4	5.1	5.3
Tb	0.4	0.6	1.1	0.5	0.4	0.9	1.1	1.1	0.8	1.6	1.2	0.7	0.7
Dy	2.5	3.9	7.3	3.2	2.5	5	6.5	5.7	4.2	8.6	7.1	4.4	4.5
Ho	0.5	0.7	1.5	0.6	0.5	0.9	1.2	1.1	0.8	1.6	1.4	0.9	0.8
Er	1.5	2.3	4.6	1.9	1.4	2.5	3.2	2.8	1.9	5.1	4.5	2.5	2.6
Tm	0.22	0.34	0.75	0.28	0.2	0.33	0.43	0.39	0.28	0.77	0.72	0.4	0.4
Yb	1.5	2.3	5	1.9	1.3	2.3	2.8	2.4	1.8	5	5.3	2.7	2.6
Lu	0.26	0.36	0.81	0.3	0.22	0.37	0.44	0.33	0.25	0.76	0.86	0.46	0.44
Hf	3	5	6.1	1.2	0.8	6.5	4.8	4	5	7	9	9.4	8.3
Ta	0.5	0.5	0.6	0.1	<0.1	1.8	2.5	2	1.4	1.5	1	1.7	2.6
Tl	0.3	0.2	0.2	<0.1	<0.1	0.2	0.2	0.5	0.1	0.9	0.7	0.4	0.5
Pb	10	5	8	<5	<5	12	7	<5	12	20	25	20	13
Bi	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4
Th	12.4	13.7	3.2	0.3	0.1	11.9	7.9	15.7	6.7	23.4	29.3	15	28.5
U	2	3.8	1.2	0.2	<0.1	2.7	2.1	3.8	1.9	5.4	13.1	3.3	9.9

Location	Kon Mom			Koh Nheak			Andong Meas			Oyadav			Svay Leu	Ph. Soporkaley	
Sample No.	CG212	CG213	CG214	CG215	CG216	CG217	CG218	CG219	CG220	CG221	CG222	CG412	CG413	CG414	CG415
SiO_2(wt %)	56.97	60.69	56.13	46.62	46.43	47.44	67.95	67.59	65.74	66.56	65.03	70.86	65.39	66.07	64.96
Al_2O_3	17.19	16.58	17.2	20.82	20.14	19.72	16.1	16.61	15.79	15.61	15.74	14.38	15.55	15.26	15.12
Fe_2O_3(T)	7.15	5.74	7.06	10.05	10.9	10.23	3.84	4.05	4.47	3.87	4.59	2.5	4.61	4.56	4.91
MnO	0.139	0.102	0.13	0.162	0.155	0.166	0.066	0.053	0.103	0.089	0.089	0.057	0.08	0.078	0.062
MgO	3.1	2.43	3.31	3.94	4.19	5.88	1.3	1.06	1.71	1.57	1.89	1.17	2.28	2.24	2.31
CaO	6.06	4.63	6.16	10.86	10.94	11.66	4.65	4.45	3.8	3.72	4.04	3.04	5.46	4.8	4.48
Na_2O	3.8	3.91	3.76	2.59	2.66	2.5	3.45	3.45	3.98	3.53	3.53	3.81	3.75	3.3	3.33

(*continued on next page*)

Table 2 (*continued*)

Location	Kon Mom			Koh Nheak			Andong Meas			Oyadav			Svay Leu	Ph. Soporkaley	
Sample No.	CG212	CG213	CG214	CG215	CG216	CG217	CG218	CG219	CG220	CG221	CG222	CG412	CG413	CG414	CG415
K_2O	3.81	4.4	3.63	1.31	1.37	1.13	1.33	2.51	3.72	3.74	3.67	2.13	0.92	2.45	2.44
TiO_2	0.98	0.885	1.016	1.411	1.503	1.061	0.364	0.335	0.541	0.511	0.587	0.257	0.454	0.594	0.621
P_2O_5	0.35	0.29	0.39	0.14	0.14	0.33	0.25	0.19	0.17	0.15	0.19	0.06	0.11	0.12	0.11
LOI	0.51	0.36	0.29	2.03	2.19	0	0	0	0	0	0	0.89	1.29	0.6	1.52
Total	100.1	100	99.07	99.93	100.6	100.1	99.3	100.3	100	99.35	99.35	99.16	99.89	100.1	99.85
Sc(ppm)	18	13	16	26	29	26	6	5	12	10	12	6	11	12	12
Be	3	3	2	<1	<1	<1	2	2	2	2	2	1	<1	1	1
V	146	111	144	318	361	317	71	50	99	84	103	48	95	102	105
Ba	466	505	482	415	392	313	631	1032	704	723	686	386	188	343	352
Sr	500	460	517	843	816	806	1024	884	462	478	467	308	387	257	254
Y	24	20	20	17	15	12	15	19	19	15	20	9	19	23	23
Zr	244	351	268	57	49	37	131	142	178	147	188	93	141	188	200
Cr	50	50	70	<20	<20	70	<20	<20	<20	<20	<20	<20	40	60	60
Ni	30	30	40	30	<20	40	<20	<20	<20	20	<20	<20	30	30	40
Cu	40	50	30	140	230	60	90	110	20	20	10	30	<10	30	20
Zn	70	60	80	70	60	60	40	<30	40	50	40	<30	30	60	40
Ga	19	19	20	19	19	18	18	17	16	16	16	14	17	15	16
Ge	1	1	1	1	1	1	<1	<1	1	1	1	1	1	1	1
As	<5	6	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5
Rb	168	245	163	28	30	20	65	70	109	116	114	37	20	114	100
Nb	17	25	16	4	4	2	6	7	6	5	6	2	2	4	5
Mo	3	<2	3	<2	<2	<2	<2	9	2	<2	3	<2	<2	<2	<2
Ag	0.5	0.8	0.7	<0.5	<0.5	<0.5	<0.5	<0.5	0.5	<0.5	<0.5	<0.5	<0.5	<0.5	0.6
In	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2
Sn	2	2	2	<1	<1	1	<1	<1	1	<1	1	<1	<1	1	1
Sb	<0.5	0.9	<0.5	<0.5	0.8	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5
Cs	11.6	24.8	13.1	1.4	1.7	1.2	1.6	0.9	1.7	2	1.9	0.7	0.9	8.9	3.6
La	29	31.3	28.5	8.2	7.8	7.8	23.7	38.2	25.4	24.8	32.3	13.2	13.8	18.1	17.4
Ce	61.3	64.1	61.2	18.5	15.6	16.8	47.1	73	53.4	44.8	60.7	24.6	29.4	39	37.9
Pr	7.37	7.46	7.27	2.54	2.2	2.25	5.25	7.87	6.38	4.81	6.72	2.63	3.7	4.72	4.71
Nd	29.8	29.2	29.1	12.5	10.4	10.4	19.7	29.4	23.9	18.3	24.9	8.7	14.4	18.8	18.9
Sm	6.4	5.6	6	3.7	3.2	2.9	3.7	5.1	4.6	3.5	4.9	1.7	3.3	4.3	4.4
Eu	1.53	1.26	1.54	1.22	1.18	1.16	1.18	1.33	1.19	1.02	1.17	0.47	1.01	0.96	0.9
Gd	5.6	4.4	5.1	4.1	3.5	3	3	3.8	3.9	3	3.9	1.5	3.4	4.1	4.2
Tb	0.8	0.7	0.7	0.6	0.6	0.4	0.4	0.5	0.6	0.4	0.6	0.3	0.6	0.7	0.7
Dy	4.9	4	4.2	3.7	3.3	2.6	2.5	3	3.4	2.5	3.4	1.6	3.5	4.1	4.2
Ho	0.9	0.7	0.8	0.7	0.6	0.5	0.5	0.6	0.7	0.5	0.7	0.3	0.7	0.8	0.8
Er	2.8	2.2	2.4	1.9	1.8	1.3	1.5	1.8	2	1.5	2	0.9	2.1	2.4	2.4
Tm	0.4	0.32	0.32	0.28	0.25	0.18	0.21	0.28	0.3	0.23	0.31	0.16	0.33	0.34	0.36
Yb	2.7	2.4	2.3	1.7	1.6	1.1	1.6	1.9	2.1	1.6	2.1	1.1	2.2	2.3	2.4
Lu	0.43	0.36	0.36	0.26	0.25	0.17	0.25	0.33	0.35	0.26	0.36	0.19	0.37	0.36	0.38
Hf	5.9	8.2	6.2	1.8	1.5	1.1	3.1	3.5	4.4	3.5	4.6	2.3	3.3	4.5	4.9
Ta	1.7	2.1	1.3	0.3	0.3	0.2	0.8	0.9	0.7	0.5	0.8	0.6	0.4	0.6	0.5
Tl	0.4	0.7	0.4	0.2	0.2	0.1	0.5	0.4	0.6	0.5	0.5	0.2	0.2	0.4	0.4
Pb	15	15	14	6	6	6	<5	7	11	12	10	<5	<5	15	9
Bi	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	0.4	<0.4	<0.4	<0.4	<0.4	<0.4
Th	20.2	32.6	15.3	1.6	1	0.9	8.7	13.8	14.4	12.4	17.4	6.7	2.5	10.6	10.4
U	5.4	7.8	6.9	0.3	0.3	0.3	1.1	2.4	4	3.5	4.6	2.4	1.1	2.8	3.1

Location	Ph. Basset North				Ph. Basset South		Dar				Ph. Aural North	
Sample No.	CG301	CG302	CG303	CG304	CG305	CG306	CG307	CG308	CG309	CG310	CG311	CG312
SiO_2(wt %)	73.55	76.1	74.74	75.52	72.55	72.78	74.9	74.29	73.03	75.1	74.95	74.66
Al_2O_3	13.27	12.51	12.92	12.11	14.07	12.89	13.26	12.51	13.91	12.43	13.05	12.64
Fe_2O_3(T)	1.84	1.82	1.61	1.49	2.48	2.83	1.56	1.46	1.88	1.32	1.11	1.7
MnO	0.053	0.052	0.049	0.041	0.061	0.067	0.086	0.077	0.07	0.047	0.029	0.025
MgO	0.32	0.28	0.26	0.22	0.47	0.49	0.13	0.1	0.31	0.09	0.12	0.12
CaO	0.81	0.96	1.01	0.79	1.82	1.41	0.59	0.7	1.07	0.33	0.85	0.42
Na_2O	3.12	3.01	3.31	2.89	3.79	3.2	3.72	3.79	3.72	3.92	3.55	3.06
K_2O	4.83	4.57	4.5	5.1	3.81	3.86	4.76	4.73	4.69	5.03	4.82	4.58
TiO_2	0.202	0.19	0.166	0.149	0.302	0.342	0.123	0.103	0.183	0.095	0.088	0.108
P_2O_5	0.07	0.05	0.05	0.05	0.1	0.15	0.03	0.04	0.04	<0.01	0.02	<0.01
LOI	0.72	0.74	0.71	0.71	0.66	0.6	1.18	1.07	0.76	0.86	0.61	1.08
Total	98.79	100.3	99.32	99.06	100.1	98.61	100.3	98.87	99.67	99.21	99.21	98.39
Sc(ppm)	3	3	3	3	4	4	4	4	4	4	3	6
Be	6	7	8	8	8	7	3	3	2	2	3	1
V	12	9	9	9	18	21	<5	<5	12	<5	7	<5

(*continued on next page*)

Table 2 (*continued*)

Location	Ph. Basset North				Ph. Basset South		Dar				Ph. Aural North	
Sample No.	CG301	CG302	CG303	CG304	CG305	CG306	CG307	CG308	CG309	CG310	CG311	CG312
Ba	138	94	80	117	121	111	303	241	571	252	316	685
Sr	44	35	33	34	81	65	40	33	94	40	74	63
Y	30	43	31	29	34	30	32	26	23	18	27	32
Zr	115	127	89	94	149	155	160	142	165	116	86	137
Cr	<20	<20	<20	<20	<20	<20	<20	<20	<20	<20	<20	<20
Ni	<20	<20	<20	30	20	<20	60	<20	<20	<20	<20	<20
Cu	20	<10	30	20	<10	<10	<10	<10	<10	<10	<10	<10
Zn	40	40	40	30	50	50	110	30	50	<30	<30	60
Ga	18	18	17	17	21	19	15	14	16	13	14	16
Ge	2	2	2	2	2	2	2	2	1	1	2	1
As	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5
Rb	358	326	316	346	292	369	203	221	208	183	256	140
Nb	15	19	14	14	20	23	11	11	10	6	5	5
Mo	10	<2	<2	52	<2	<2	<2	<2	<2	<2	<2	<2
Ag	<0.5	<0.5	<0.5	<0.5	0.6	<0.5	0.7	<0.5	0.7	<0.5	<0.5	<0.5
In	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2
Sn	11	17	9	11	14	18	6	2	2	1	3	2
Sb	0.5	<0.5	<0.5	<0.5	<0.5	<0.5	0.9	<0.5	<0.5	<0.5	<0.5	<0.5
Cs	21.8	20.5	14.1	12.3	28.9	33.8	6.8	8.3	11.1	6.6	11.5	3
La	25.4	27.4	18.2	22.3	19.2	34.2	63.2	45.6	40.6	36.7	27.7	38
Ce	52.6	54.7	37.7	45.5	38.8	66.3	103	90	78.7	66.4	53.8	77.5
Pr	5.98	6.02	4.28	5	4.4	7.1	12.9	9.43	8.09	7.49	5.93	9.27
Nd	20.8	20.9	15.3	17.1	16.3	25.1	43	32.8	27.4	25.5	21.1	33
Sm	4.8	5	3.9	4.2	4.5	5.3	7.9	6.4	5.2	4.6	4.6	6.6
Eu	0.38	0.32	0.28	0.28	0.5	0.51	0.58	0.45	0.68	0.46	0.41	0.59
Gd	4.3	5.1	3.9	4.1	4.7	5.2	6.6	5.2	4.5	3.9	4.3	5.5
Tb	0.9	1	0.8	0.8	1	0.9	1	0.9	0.7	0.6	0.8	0.9
Dy	5.2	6.9	4.8	5	6.2	5.5	5.9	5.3	4.2	3.3	4.9	5.3
Ho	1	1.4	1	1	1.2	1.1	1.2	1	0.8	0.7	1	1
Er	3.1	4.5	3.2	3.1	3.6	3.1	3.6	3	2.7	2.1	3.1	2.9
Tm	0.53	0.8	0.51	0.49	0.57	0.49	0.55	0.46	0.38	0.31	0.47	0.45
Yb	3.7	5.7	3.7	3.5	3.9	3.7	3.3	3.4	2.8	2.1	3.3	3
Lu	0.57	0.95	0.59	0.56	0.58	0.6	0.52	0.51	0.45	0.32	0.52	0.48
Hf	3.3	3.9	2.7	2.8	4	4.1	4.1	3.7	4	2.9	2.6	3.5
Ta	4.7	8.1	5	6	4.5	6.4	1.5	1.8	1.4	1.1	1.6	1
Tl	1.4	1.5	1.4	1.8	1.4	1.4	1.2	1.2	1.2	0.8	1	0.6
Pb	34	34	34	34	27	31	89	57	36	73	31	23
Bi	<0.4	0.5	1.7	0.6	0.8	<0.4	<0.4	1	<0.4	<0.4	<0.4	<0.4
Th	33.2	35.1	29.7	30.5	24.4	32.9	26.7	30.6	26	22.8	19.8	14.2
U	15.7	17.7	16	17.6	13	11.4	6.7	6	4.8	4.3	7.6	2.2

Location	Ph. Aural North	Ph. Aural South			Chum Kiri		Kirivong		Ph. Tamao			Ba Phnom
Sample No.	CG314	CG315	CG316	CG317	CG318	CG319	CG320	CG321	CG322	CG323	CG401	CG402
SiO_2(wt %)	77.5	70.96	69.58	71.21	73.34	65.97	73.55	74.05	76.71	69.06	74.15	75.41
Al_2O_3	12.12	14.36	15.65	13.99	13.99	15.95	13.88	13.75	11.7	14.41	12.93	12.72
Fe_2O_3(T)	0.67	2.77	3.18	3.27	1.62	4.74	1.28	1.59	1.84	5.35	2.05	1.83
MnO	0.008	0.049	0.057	0.053	0.055	0.072	0.014	0.065	0.053	0.102	0.032	0.027
MgO	0.03	0.38	0.65	0.51	0.56	0.44	0.08	0.08	0.24	0.76	0.24	0.16
CaO	0.52	2.02	3.32	2.36	0.56	2.13	1.29	0.22	0.9	1.6	1.31	1.05
Na_2O	3.43	3.81	4.13	3.43	3.81	4.67	3.83	3.74	3.19	3.78	3.31	3.22
K_2O	4.46	3.09	2.57	3.56	3.9	3.2	4.73	4.52	3.9	3.97	4.8	4.82
TiO_2	0.056	0.254	0.351	0.353	0.188	0.512	0.13	0.075	0.186	0.579	0.224	0.189
P_2O_5	<0.01	0.08	0.09	0.09	0.04	0.1	0.02	0.03	0.06	0.24	0.03	0.03
LOI	0.54	0.87	0.83	0.54	1.58	1.25	1.03	1.09	1.24	0.62	0.71	0.62
Total	99.33	98.65	100.4	99.37	99.65	99.03	99.83	99.22	100	100.5	99.77	100.1
Sc(ppm)	3	5	6	9	4	13	4	4	4	10	4	4
Be	2	3	3	2	2	7	5	2	6	7	2	3
V	<5	17	30	27	13	23	8	<5	10	28	10	8
Ba	153	257	269	539	766	189	153	420	83	106	605	406
Sr	26	136	205	197	97	105	57	54	36	51	67	47
Y	31	33	23	23	18	166	103	18	39	79	44	50
Zr	84	156	173	173	102	454	160	82	133	398	188	173
Cr	<20	90	<20	<20	<20	<20	<20	<20	<20	<20	<20	<20
Ni	<20	<20	<20	<20	<20	<20	<20	<20	<20	<20	30	<20
Cu	20	<10	<10	<10	<10	<10	<10	<10	<10	<10	10	<10
Zn	<30	50	60	420	<30	110	<30	220	50	120	50	50

(*continued on next page*)

Table 2 (*continued*)

Location	Ph. Aural North	Ph. Aural South			Chum Kiri		Kirivong		Ph. Tamao			Ba Phnom
Sample No.	CG314	CG315	CG316	CG317	CG318	CG319	CG320	CG321	CG322	CG323	CG401	CG402
Ga	15	19	20	17	14	35	26	18	18	30	19	19
Ge	2	2	1	1	1	2	2	1	2	2	1	2
As	<5	<5	<5	<5	<5	7	5	38	<5	<5	<5	<5
Rb	231	169	133	143	126	393	400	186	291	431	228	243
Nb	8	7	7	6	6	42	18	10	17	48	4	5
Mo	<2	<2	<2	<2	4	2	<2	4	<2	3	<2	<2
Ag	<0.5	<0.5	0.6	0.6	<0.5	1.5	0.6	<0.5	<0.5	1.2	<0.5	0.6
In	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2
Sn	<1	4	3	2	1	7	14	5	12	27	5	4
Sb	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5
Cs	3.9	10.5	5.2	5.4	2.9	16.9	10.1	3.1	16.6	42.3	16.4	11.5
La	34.2	44.4	33.9	32	27.4	155	49.6	21.3	26.5	130	40.7	45.8
Ce	60.1	74.5	63.9	65	51	347	110	43.7	54.6	256	82.5	93
Pr	8.09	9.34	6.94	7.11	5.14	40.9	13.1	4.77	6.07	27.2	9.44	10.5
Nd	28.3	32.3	24	26.8	17.1	149	49.2	15.9	21.3	91.4	35.2	38.6
Sm	5.7	6.3	4.7	5.5	3.2	34	12.6	3.5	5.4	19.4	7.4	8.2
Eu	0.37	0.7	0.88	0.85	0.58	0.7	0.41	0.34	0.26	0.41	0.75	0.53
Gd	5.1	5.5	4.2	4.9	2.8	31.3	13.3	3.2	5.4	17.1	7	7.9
Tb	0.9	0.9	0.7	0.8	0.5	5.5	2.6	0.6	1.1	3	1.2	1.4
Dy	5.4	5.8	4.2	4.7	3	33.7	17	3.2	6.6	15.7	7.8	8.8
Ho	1.1	1.2	0.9	0.9	0.6	6.4	3.3	0.6	1.3	2.8	1.5	1.7
Er	3.6	3.6	2.6	2.5	1.9	18.1	10.3	1.8	3.9	7.5	4.5	5.3
Tm	0.56	0.53	0.38	0.37	0.29	2.59	1.66	0.26	0.62	1.1	0.63	0.81
Yb	3.9	3.9	2.5	2.4	2.1	16.2	10.9	1.8	4	7.2	4	5.1
Lu	0.63	0.58	0.4	0.36	0.36	2.37	1.64	0.27	0.58	1.05	0.59	0.76
Hf	3.3	4	4.3	4.1	2.3	13.5	5.5	2.3	3.6	10.3	4.8	5.4
Ta	1.9	1.2	1.3	0.8	1.2	3.9	3.6	1.4	4.2	7	0.9	1.4
Tl	0.8	0.7	0.6	0.6	0.6	1.8	2	1.1	1.2	2.1	1.2	0.8
Pb	25	23	20	22	14	28	31	35	32	29	26	27
Bi	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	0.7	0.7	0.8	<0.4	<0.4
Th	28.1	22.1	21.1	11.8	10.6	82.7	34.9	13	32.8	108	18.9	24.3
U	6.6	5.1	5.8	1.9	2.1	16.2	11.4	3.5	12.6	17.4	4.6	6.6

Location	Ba Phnom			Ph. Da			Ph. Rab Bat		Ratanakmundol	
Sample No.	CG403	CG404	CG405	CG406	CG407	CG408	CG409	CG410	CG411A	C-G411B
SiO_2(wt %)	73.68	78.7	78.59	77.21	76.36	75.43	74.01	74.12	78.69	77.61
Al_2O_3	13.51	10.72	11.06	12.35	12.69	12.53	13.3	13.11	11.66	11.35
Fe_2O_3(T)	2.35	0.94	0.92	0.93	0.89	1.18	1.62	1.42	0.54	1.31
MnO	0.031	0.012	0.009	0.03	0.05	0.03	0.048	0.042	0.009	0.014
MgO	0.25	0.02	0.02	0.03	0.02	0.08	0.1	0.09	0.12	0.13
CaO	1.34	0.33	0.44	0.5	0.5	0.48	0.96	0.9	1.16	1.21
Na_2O	3.56	3.4	3.52	3.71	4.2	3.47	3.93	3.75	5.81	5.86
K_2O	4.3	4.2	4.46	4.79	4.12	4.75	4.4	4.61	0.14	0.14
TiO_2	0.279	0.06	0.062	0.042	0.025	0.077	0.099	0.091	0.144	0.134
P_2O_5	0.04	<0.01	<0.01	<0.01	<0.01	<0.01	0.02	0.02	<0.01	<0.01
LOI	0.81	0.34	0.5	0.6	0.42	0.61	0.77	0.64	1.64	1.71
Total	100.2	98.7	99.59	100.2	99.29	98.64	99.26	98.79	99.91	99.48
Sc(ppm)	5	2	1	3	3	2	3	3	12	13
Be	3	4	4	5	6	6	3	2	<1	<1
V	11	<5	<5	<5	<5	5	<5	<5	<5	<5
Ba	444	65	114	33	6	56	464	523	50	48
Sr	73	5	8	12	4	23	69	68	51	24
Y	217	82	69	117	163	89	24	21	78	71
Zr	242	90	143	101	107	102	136	125	173	156
Cr	<20	<20	<20	<20	<20	<20	<20	<20	<20	<20
Ni	<20	<20	<20	<20	<20	30	<20	<20	<20	<20
Cu	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10
Zn	60	90	50	30	<30	<30	40	30	<30	<30
Ga	20	22	23	24	26	22	18	17	13	12
Ge	2	2	2	2	3	2	2	1	1	1
As	<5	<5	<5	<5	<5	<5	<5	<5	<5	<5
Rb	207	234	240	461	685	625	192	194	<2	<2
Nb	6	8	9	25	35	16	9	8	<1	1
Mo	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2
Ag	0.7	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5
In	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2

(*continued on next page*)

Table 2 (*continued*)

Location	Ba Phnom			Ph. Da			Ph. Rab Bat		Ratanakmundol	
Sample No.	CG403	CG404	CG405	CG406	CG407	CG408	CG409	CG410	CG411A	C-G411B
Sn	6	5	<1	3	9	16	2	2	2	<1
Sb	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5	<0.5
Cs	8.2	7.4	3.7	4	8.7	13.7	6	6.2	<0.5	<0.5
La	159	19.3	32	22.9	10.2	24.6	36.2	29.9	18	7.9
Ce	303	49.2	69.5	58.9	30.9	56.2	70.6	59.9	48.7	22.2
Pr	38.8	6.72	8.38	7.35	4.49	6.84	7.52	6.54	7.12	3.45
Nd	152	29.1	32.2	28.7	17.9	24.9	26.4	23.2	33.4	17.7
Sm	36.9	9.4	8.4	10	8.7	7.1	5.3	4.6	9.7	6
Eu	2.15	0.19	0.27	0.12	<0.05	0.27	0.61	0.58	0.89	0.67
Gd	38.4	11.5	9.4	10.3	10.7	7.8	4.4	3.6	10.6	8.4
Tb	6.5	2.2	1.7	2.4	2.7	1.7	0.7	0.6	2	1.7
Dy	38.1	14.3	11.3	16.2	20.1	11.7	4.2	3.7	13.2	12
Ho	7.2	2.9	2.4	3.4	4.5	2.6	0.8	0.7	2.8	2.7
Er	18.7	8.4	6.9	10.9	15.7	8.4	2.5	2.1	8.4	8.2
Tm	2.54	1.22	1.03	1.87	2.86	1.4	0.36	0.33	1.33	1.37
Yb	15.2	7.3	6.5	13	21	10	2.6	2.1	8.8	9.6
Lu	2.07	0.98	0.93	2.02	3.22	1.57	0.39	0.33	1.38	1.45
Hf	6.5	3.9	5.3	5	6.7	4.1	3.9	3.5	5.1	4.8
Ta	0.9	1.7	1.5	8	14	6.9	1.6	1.4	0.4	0.4
Tl	1.2	1.2	1	1.9	2.8	3.3	1.2	1	0.2	0.1
Pb	23	33	23	55	62	46	25	22	<5	<5
Bi	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4	<0.4
Th	19.7	20.5	21.3	39.7	36.8	43.4	25	21.9	1.5	1.4
U	5.7	4.2	5.4	17	25.2	14.4	5.5	5	0.6	0.7

Cambodia on the Sr/Y vs. Y diagram, which is useful for identifying adakitic rocks.

In the NE region of Cambodia, some igneous rocks (Andong Meas, Koh Nheak, and part of Phnom Daek, Phnom Koy Rmeas, Svay Leu, Oyadav, and Svay Chras (CG208)) are classified as adakitic rocks, but other rocks in the NE region are non-adakitic rocks. All the igneous rocks in the SW region are non-adakitic, and some A-type granitic rocks in the SW region (Kirivong, Phnom Da, and Ba Phnom) have very high Y.

4.2.6. REE patterns and Harker diagrams for incompatible elements

Chondrite-normalized REE patterns have been used to illustrate the behaviors of rare earth elements (REEs) (McDonough and Sun, 1995). In the REE patterns, europium (Eu) anomalies are important for identifying plagioclase fractionation in igneous magma. Europium is unique among REEs in having a reduced Eu^{2+} state in addition to an oxidized Eu^{3+} state. Under relatively reduced conditions, Eu exists mainly as Eu^{2+}, whereas the other REEs exist as REE^{3+}. Eu^{2+} has a lower charge and a larger ionic radius, similar to Sr^{2+} and Ca^{2+}. As with Sr^{2+}, Eu^{2+} substitutes preferentially for Ca^{2+} in plagioclase. Therefore, the fractionation of plagioclase will make a magma depleted in Eu relative to the other REEs (Gill, 2010). In addition, because adakitic magma forms in deep locations in coexistence with HREE-rich garnet, it is depleted in HREEs. As a result, the LREEs/HREEs content increases in the adakitic magma. In contrast, A-type granitic rocks have high REE content and incompatible elements, but are low in Ba, Sr, and Eu.

The REE patterns of the igneous rocks in Cambodia differ greatly between the NE and SW regions (Fig. 12). The igneous rocks in the NE region, except for gabbro in Phnom Koy Rmeas and Koh Nheak are low in HREEs and high in LREEs/HREEs. The gabbro in Phnom Koy Rmeas and Koh Nheak shows a flat pattern (M-type) (Fig. 12). Most igneous rocks in the NE region do not show a negative Eu anomaly, but some igneous rocks in Kon Mom, Svay Chras, and Phnom Soporkaley show a slightly negative Eu anomaly. In contrast, the igneous rocks in the SW region show a negative Eu anomaly. This negative Eu anomaly is prominent in the A-type granitic rocks in Kirivong, Phnom Tamao, Ba Phnom, and Phnom Da, which are enriched in REEs, but is moderate for the I-type and I-S-type (ilmenite series) granitic rocks (Dar, Phnom Rab Bat, Phnom Aural North and South, Phnom Baseth North and South, Chumkiri, and Ratanakmundol).

Harker diagrams of Rb, Y, Sn, Pb, Th, and U vs. SiO_2 are useful to evaluate the influence of contamination from the continental crust, which is rich in incompatible elements compared with the mantle. The igneous rocks in the NE region tend to be low in these incompatible elements compared with the ones in the SW region (Fig. 13). There is no clear difference between adakitic and non-adakitic rocks. This suggests that the igneous rocks in the NE region were subjected to only small levels of contamination from the continental crust materials and/or negligible differentiation. The igneous rocks in the SW region are more enriched in these elements than the ones in the NE region. Considering the negative Eu anomaly, the data indicate that the magma responsible for the formation of the granitic rocks in the SW region may have contained materials from the lower to middle continental crust and/or experienced plagioclase fractionation.

4.2.7. Chemical composition of biotite

The average chemical composition of biotite from the igneous rocks is summarized in Table 3. The total Al (^{T}Al) content in biotite on the basis of 22 oxygen atoms ranged from 2.2 to 3.2 in the igneous rocks in the NE region and from 2.3 to 3.5 in the igneous rocks in the SW region (Fig. 14). There was no significant difference in the ^{T}Al content in biotite between the NE and SW regions. Conversely, the Mg/(Mg + Fe) ratio in the biotite is significantly higher in the igneous rocks in the NE region (0.45–0.7) than in the igneous rocks in the SW region (0.05–0.3). This may suggest that the igneous rocks in the SW region were more differentiated than the NE igneous rocks or that the igneous rocks in the SW region were significantly affected by continental crust materials.

4.3. Origin and ages of igneous magmas: Sr and Nd isotopic evidence

4.3.1. Age determination using Rb-Sr whole rock isochron dating

A Rb-Sr isochron dating method has long been used for dating igneous rocks (Dickin, 2005). The ^{87}Sr in rocks is produced by the decay of ^{87}Rb. When a series of rocks were formed from the same source

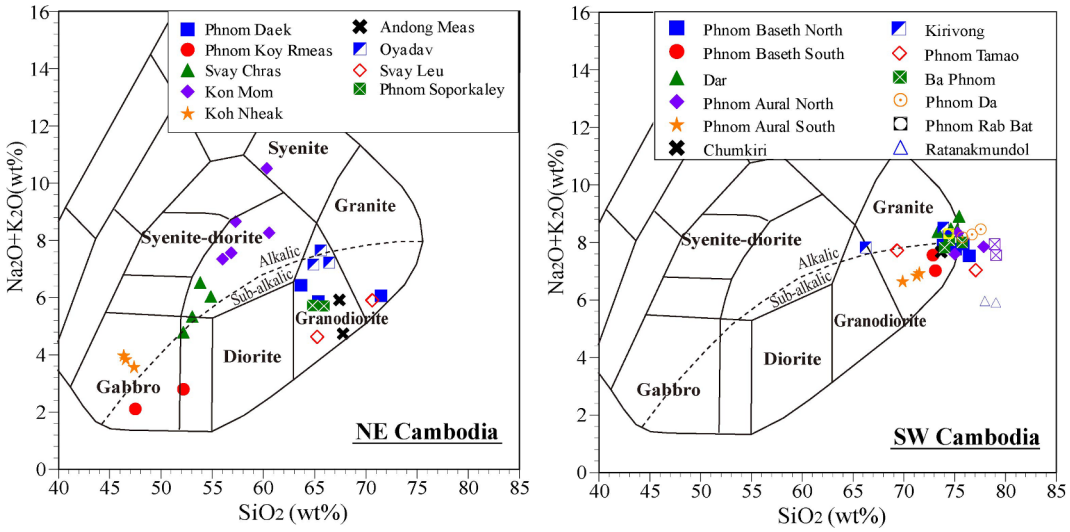


Fig. 4. Classification of the studied igneous rocks based on SiO_2 and $Na_2O + K_2O$ contents. The classification boundaries are after Cox et al. (1979) and Wilson (1989).

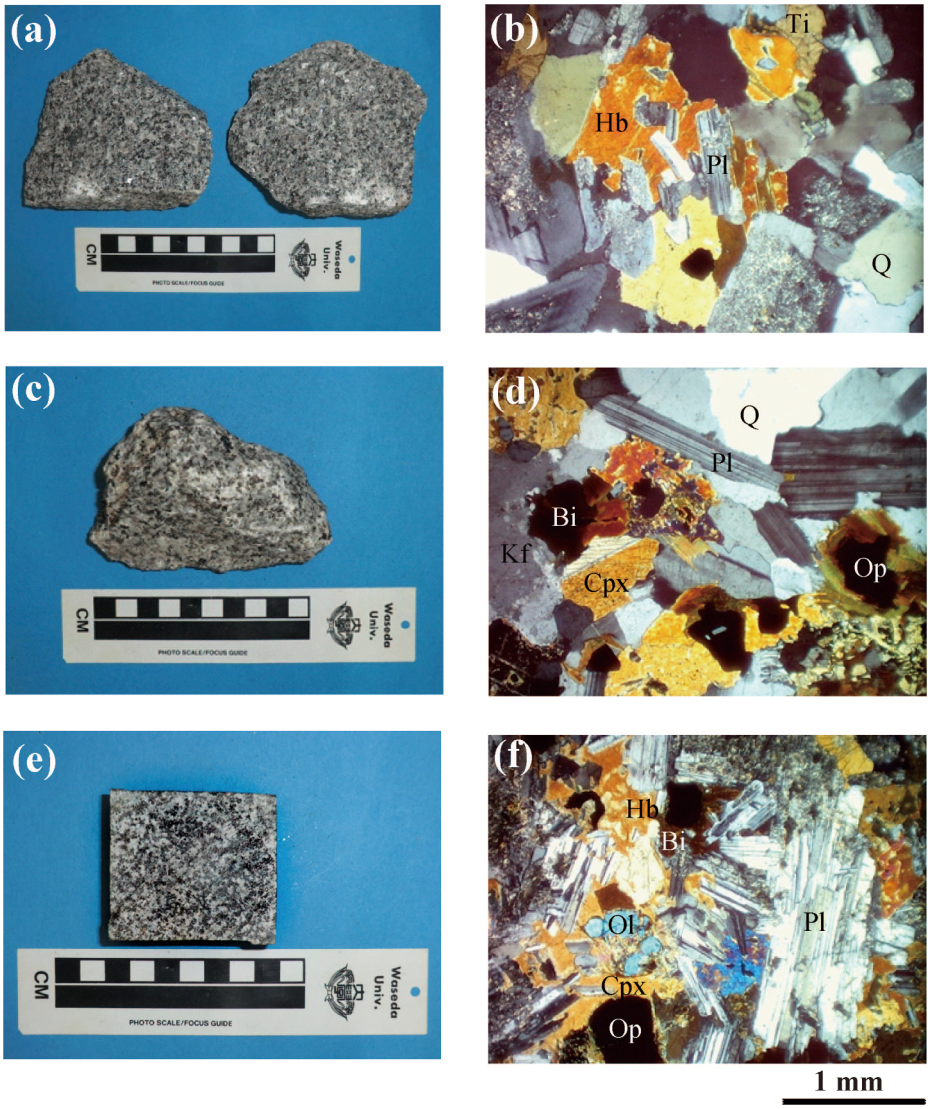


Fig. 5. Photographs and photomicrographs under crossed polars of the igneous rocks from NE Cambodia. (a) and (b) for granodiorite in Phnom Daek (sample no. CG202), (c) and (d) for syenite-diorite in Kon Mom (sample no. 214), and (e) and (f) for gabbro in Koh Nheak (sample no. 217). Abbreviations: Bi, biotite; Cpx, clinopyroxene; Hb, hornblende; Kf, potassium feldspar; Ol, olivine; Op, opaque minerals; Pl, plagioclase; Q, quartz; Ti, titanite.

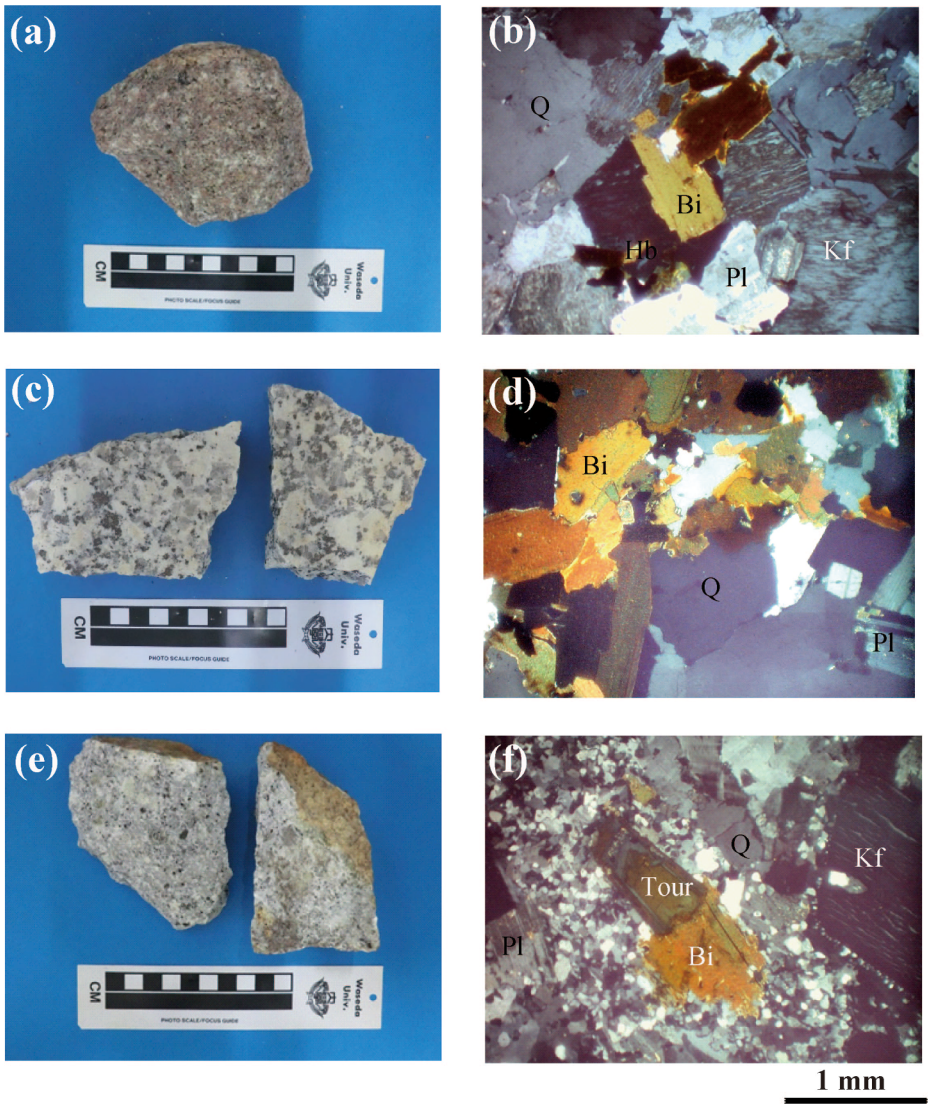


Fig. 6. Photographs and photomicrographs under crossed polars of the igneous rocks from SW Cambodia. (a) and (b) for granite in Dar (sample no. CG307), (c) and (d) for granite in Kirivong (sample no. CG3193), and (e) and (f) granite in Phnom Baseth North (sample no. CG303). Abbreviations: Bi, biotite; Hb, hornblende; Kf, potassium feldspar; Pl, plagioclase; Q, quartz; Tour, tourmaline.

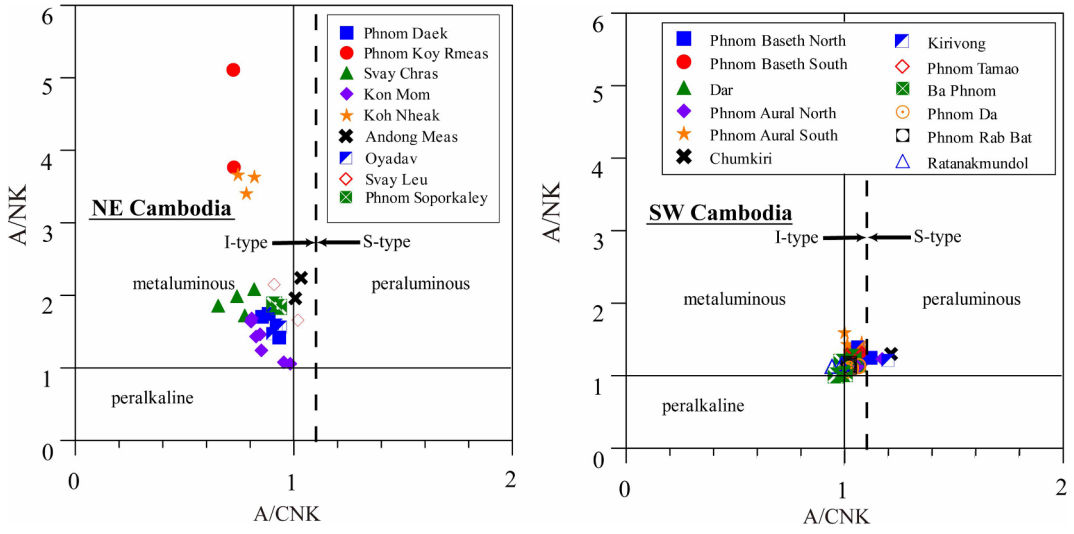


Fig. 7. $Al_2O_3/(Na_2O + K_2O)$ vs. $Al_2O_3/(CaO + Na_2O + K_2O)$ diagram showing the classification of I- and S-type igneous rocks.

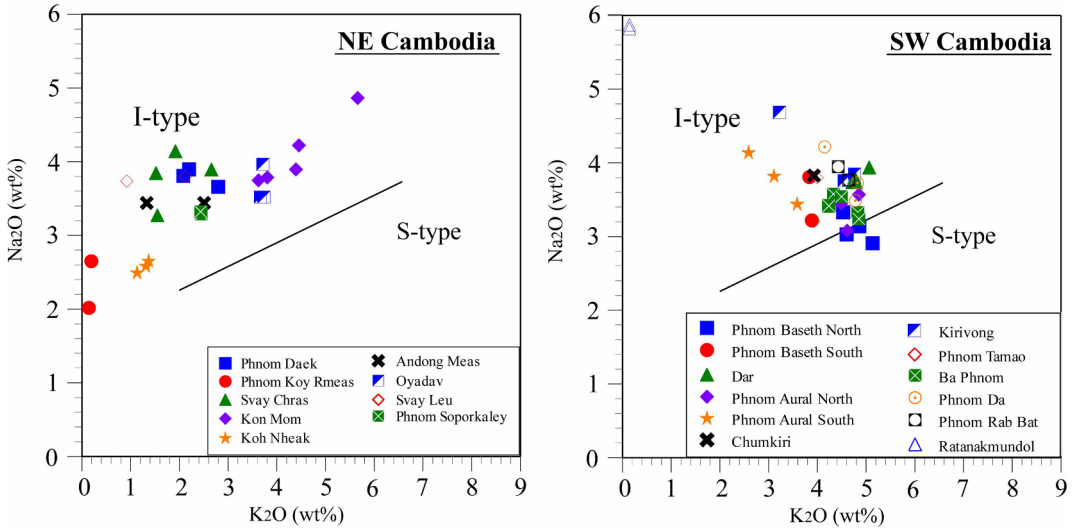


Fig. 8. Na_2O vs. K_2O diagram by Chappell and White (1974, 2001) showing the classification of I- and S-type igneous rocks.

material and have been in a closed since formation, the $^{87}Sr/^{86}Sr$ and $^{87}Rb/^{86}Sr$ of the rock at present ($(^{87}Sr/^{86}Sr)_{present}$ and $(^{87}Rb/^{86}Sr)_{present}$) are expressed as follows:

$$(^{87}Sr/^{86}Sr)_{present} = (^{87}Sr/^{86}Sr)_i + (^{87}Rb/^{86}Sr)_{present}(e^{\lambda t} - 1)$$

where λ ($=1.42 \times 10^{-11}$ yr^{-1}) is the decay constant for ^{87}Rb, t is time in years elapsed after the formation of the rock, and $(^{87}Sr/^{86}Sr)_i$ is the $^{87}Sr/^{86}Sr$ at the time of rock formation, termed the initial ratio.

Likewise, in the case of the Sm-Nd isotopic system, the following equation holds:

$$(^{143}Nd/^{144}Nd)_{present} = (^{143}Nd/^{144}Nd)_i + (^{147}Sm/^{144}Nd)_{present}(e^{\lambda t} - 1)$$

where λ, the decay constant of ^{147}Sm, is given as 6.54×10^{-12} yr^{-1}.

The Sr and Nd isotopic ratios of the igneous rocks are summarized in Table 4. We determined the formation ages of several igneous rocks with the Rb-Sr whole rock isochron method using the software IsoplotR (Vermeesch, 2019). The initial ratio of $^{143}Nd/^{144}Nd$ for igneous rocks was also calculated using the formation age determined from the Rb-Sr isochron.

A straight line was obtained on the Rb-Sr isochron diagram for several igneous rocks in the NE region from Koh Nheak, Andong Meas, Oyadav, and Svay Chras (CG208; adakitic rock) giving an isochron age of 332 ± 5 Ma (Fig. 15). The Rb-Sr whole rock isochron ages from other igneous rocks in the NE region are 274 ± 15 Ma for Phnom Daek and Phnom Koy Rmeas, 267 ± 5 Ma for Svay Leu, 118 ± 30 Ma for Phnom Soporkaley, 103 ± 2 Ma for Kon Mon, and 97 ± 14 Ma for Svay Chras (except for CG208). These results suggest that magmatic activity in the NE region ranges from the Carboniferous to the Cretaceous, and is divided into three ages: Middle Carboniferous, Late Permian, and Cretaceous.

In the SW region, granites from Phnom Baseth North and South yielded a Rb-Sr isochron age of 167 ± 8 Ma (Fig. 15). This age is consistent with Kong et al. (2011), who reported 185 ± 0.5 Ma for the Phnom Baseth North granites using the $^{40}Ar/^{39}Ar$ method. The ages of the other granites in the SW region are 190 ± 5 Ma for Dar and Phnom Rab Bat, 205 ± 5 Ma for Ba Phnom, and 198 ± 5 Ma for Phnom Aural North and South and Chumkiri. The Rb-Sr isochron ages of the Phnom Tamao granites and Kirivong granites were 182 ± 43 Ma and 71 ± 6 Ma, respectively. These ages are consistent with the U-Pb zircon age by Fyhn et al. (2016), who reported 195 Ma for Phnom Tamao, 75 Ma for Kirivong (Tonleap), and 95 Ma for Kirivong (Phnom

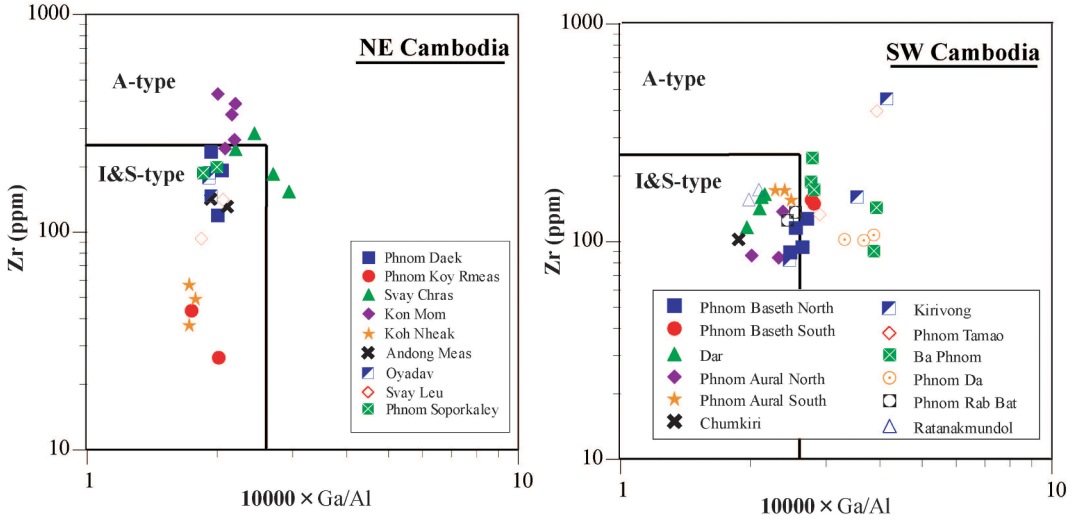


Fig. 9. I&S-type and A-type discriminant diagram for igneous rocks based on Zr and 10,000 × Ga/Al contents (Whalen et al., 1987).

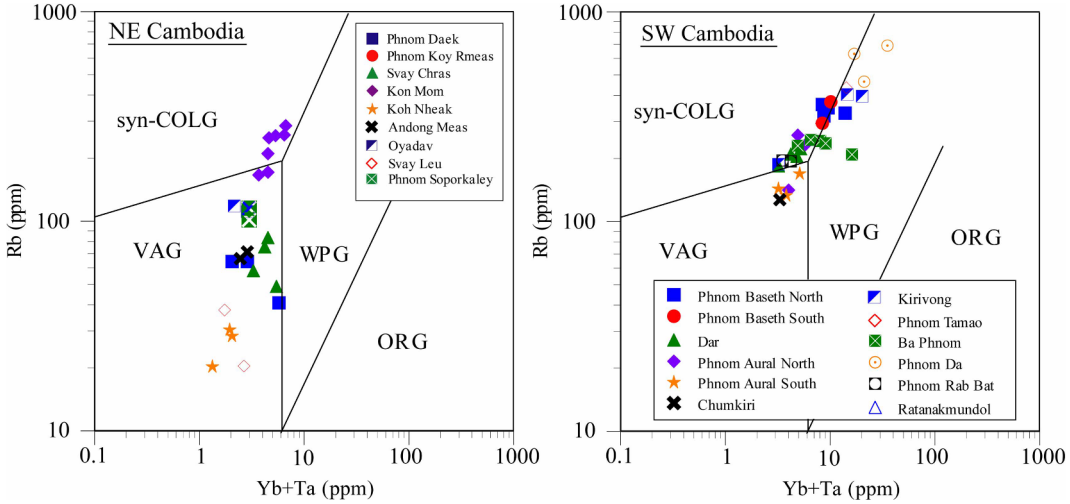


Fig. 10. Tectonic discriminant diagram for igneous rocks according to Pearce et al. (1984).

Den). The youngest age was obtained for Phnom Da granites, dated 58 ± 4 Ma. Thus, the magmatic activity of igneous rocks in the SW region is mainly Jurassic to Cretaceous, ending in the Paleogene.

4.3.2. Magma source of igneous rocks from Sr and Nd isotope ratios

The initial isotope ratios of Sr and Nd of the igneous rocks were used to provide constraints on the magma source. The following four end-members are proposed as the source of mantle-derived magma: depleted MORB mantle (DMM), enriched mantle 1 (EM1), enriched mantle 2 (EM2), and high μ (HIMU). The upper mantle, which participated in the formation of the continental crust starting prior to 4 Ga, is presumed as the source for DMM. EM1 and EM2 were derived from basalt and sediments of the oceanic crust. EM1 was affected by lower continental crust materials whereas EM2 was affected by upper continental crust materials. HIMU has high U/Pb (i.e., $^{238}U/^{204}Pb$) and Th/Pb ratios. The primitive mantle (PM) represents the average composition of Earth's crust and mantle (Schaefer, 2016). The mantle array refers to magmatic rocks of mantle origin and represents all possible mixtures of magmas, from a source resembling the depleted upper mantle to a source resembling the lower primitive mantle (Nédélec and Bouchez, 2015).

The initial isotope ratios of Sr and Nd for the igneous rocks in Cambodia clearly differed between the NE and SW regions (Fig. 16), indicating differences in source materials responsible for the formation of the two regions. The initial isotopic ratios of Sr and Nd in the igneous rocks in the NE region showed small variations concentrated around PM values, indicating a strong involvement of mantle materials. The composition of the igneous rocks in the NE region varied greatly depending on the rock species (from granitic to gabbroic rocks) and their formation ages range widely from the Carboniferous to the Cretaceous, but there is no clear difference in Sr and Nd isotopic ratios. These results indicate that melting of the mantle and/or oceanic crust strongly contributed to magma formation in the NE region over a long geologic time.

The igneous rocks in the SW region also had small variations in the initial isotope ratio of Nd (0.5123–0.5126). Because the Sm/Nd ratios of the igneous rocks in the SW region are low and fall within a small range (about 0.2–0.3) (Table 2), the increase of $^{143}Nd/^{144}Nd$ after their formation is very small, less than 0.0002 overall. However, the initial isotopic ratios of Sr (0.705–0.73) in the igneous rocks in the SW region were higher than those in the igneous rocks in the NE region and their variations were large, as were those of the Rb/Sr ratios. Further, there

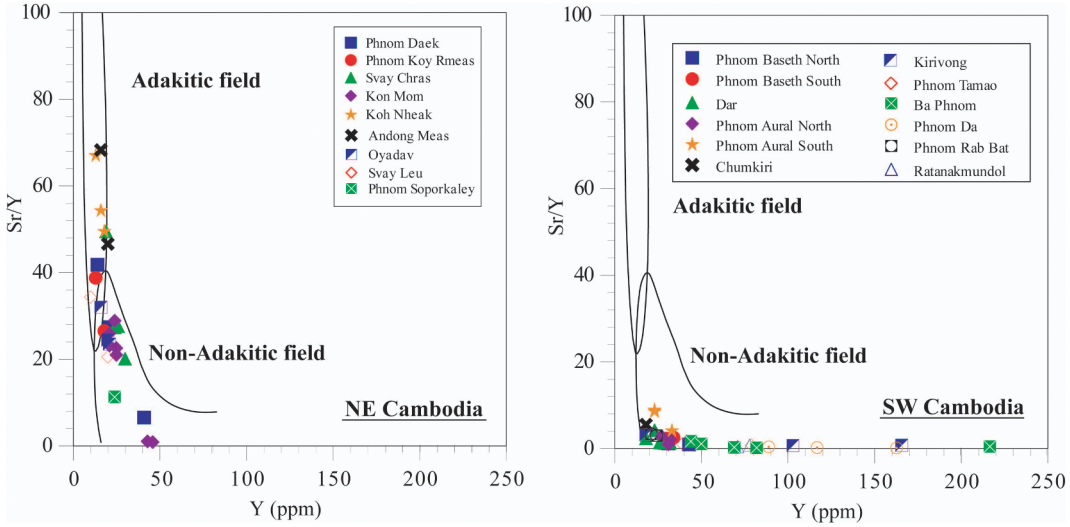


Fig. 11. Sr/Y vs. Y discriminant diagram for adakitic rocks according to Defant and Drummond (1990).

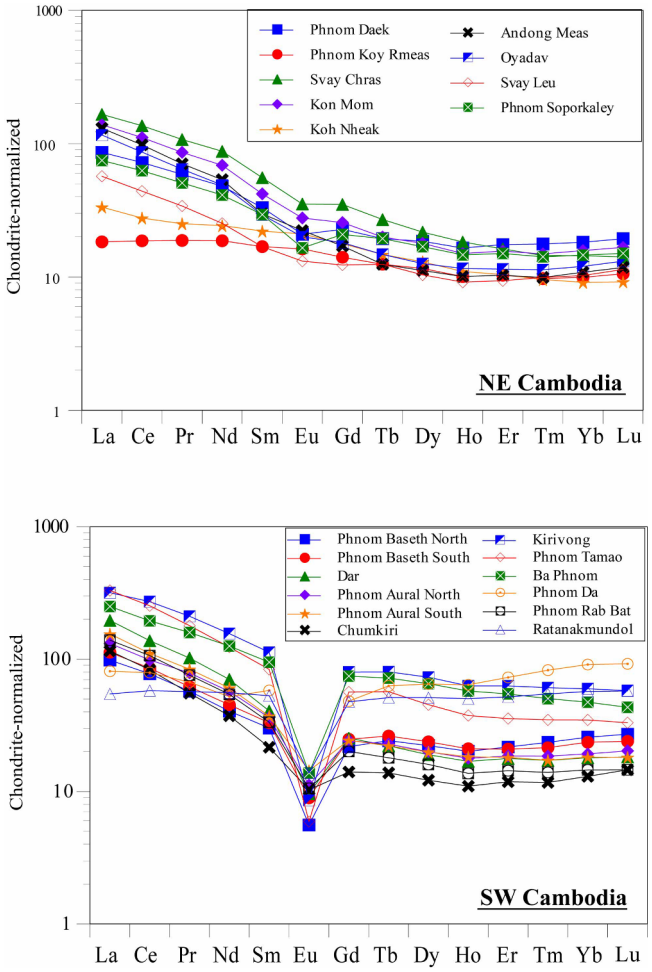


Fig. 12. Chondrite-normalized REE patterns for the igneous rocks from Cambodia.

was no negative relationship between the initial isotope ratios of Nd and Sr, suggesting a low possibility of mixing of mantle and continental crust materials with different Sr and Nd isotopic ratios. These results suggest that the igneous rocks in the SW region were formed from source materials with a relatively uniform isotope ratio of Nd. It is likely that the lower to middle continental crust was strongly involved in the formation of the igneous rocks in the SW region.

4.4. Tectonic environment of the igneous rocks

The igneous rocks in Cambodia are divided by the inferred Mae Ping fault into NE and SW regions; the formation environment, age, and source materials in the two regions are different. The adakitic rocks distributed in the NE region are divided into two groups based on formation age; the younger rocks (Phnom Daek, Phnom Koy Rmeas and Svay Leu) in the Late Permian and older rocks (Koh Nheak, Andong Meas, Oyadav, and Svay Chras (CG208)) in the Middle Carboniferous. The younger adakitic rocks may correspond to granitic rocks in the Loei fold belt located north of the Mae Ping fault in northeastern Thailand because of the similar age (Zaw et al., 2009) and chemical compositional signatures (Kamvong et al., 2014; Nualkhao et al., 2018). The older ones may correspond to those in the Po Ko Suture zone near the three-way border between Cambodia, Laos, and Vietnam, west of the Kontum block of central Vietnam (Sanematsu et al., 2011).

The Indochina Terrane includes a cryptic suture running from eastern Thailand (Saraburi) through Cambodia (Siem Reap – Preah Vihear – Stung Treng – Ratanakiri) toward the Po Ko Suture zone (Morley et al., 2013). The Indochina Terrane is divided into western and eastern parts by the north-striking Po Ko Suture zone. The old adakitic igneous rocks in NE Cambodia were formed by the west-dipping subduction and collision of the Kontum block along the Po Ko Suture zone beneath the Khorat block (Lepvrier et al., 2004, 2008).

The syenite-diorites in NE Cambodia (Svay Chras except for CG208, Kon Mom, and Phnom Soporkaley) are classified as non-adakitic and of the magnetite series, with ages from 118 Ma to 97 Ma. They may correspond to the Dinhquan granitic rocks in the Dalat-Kratie zone in Vietnam. According to Thuy et al. (2004), the

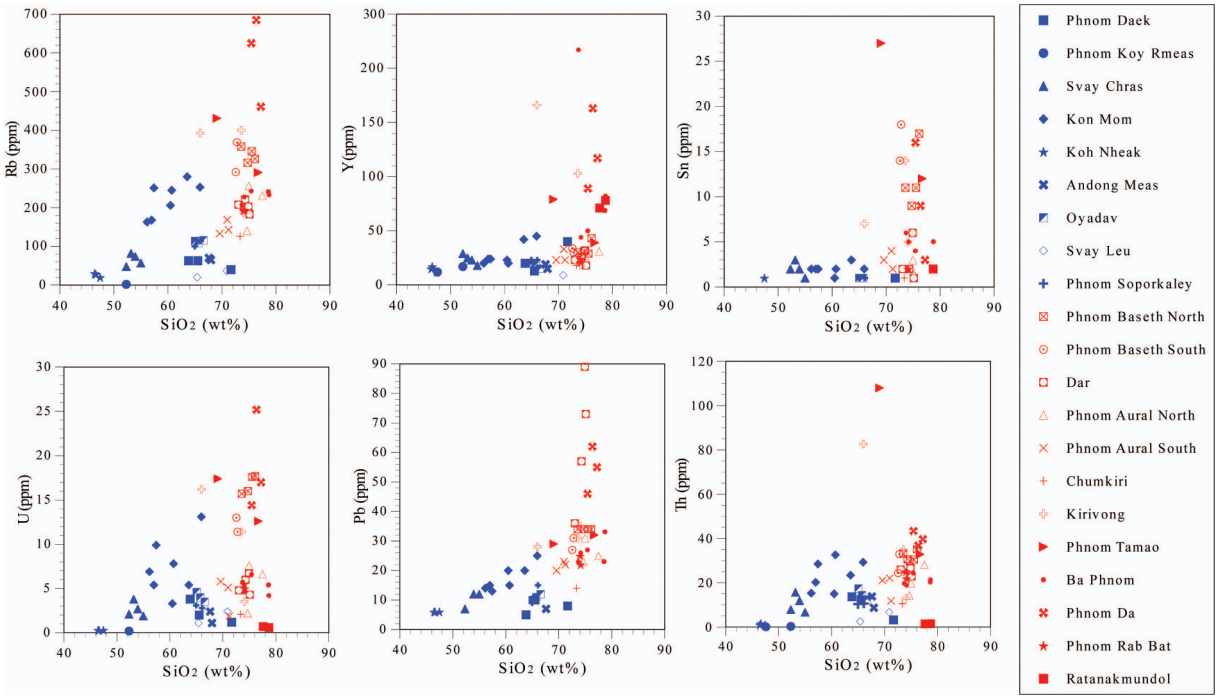


Fig. 13. Harker diagrams of incompatible elements of Rb, Y, Sn, Pb, Th, and U for the igneous rocks from Cambodia.

Table 3
Average chemical compositions of biotite from igneous rocks in Cambodia, determined by SEM-EDX.

	Phnom Daek		Svay Chras			Kon Mom				Koh Nheak	Andong Meas		Phnom Baseth North			
Sample No.	CG201	CG202	CG206	CG207A	CG207B	CG211	CG212	CG213	CG214	CG217	CG218	CG219	CG301	CG302	CG303	CG304
Number of samples	2	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
Oxide(wt%)																
SiO_2	38.12	38.08	37.86	36.61	37.14	38.98	37.29	36.85	37.79	36.16	36.71	36.65	36.03	35.78	35.85	35.39
TiO_2	4.57	4.23	5.57	5.49	5.57	6.23	5.22	4.65	5.17	4.75	3.79	3.90	3.85	3.52	3.83	3.61
Al_2O_3	13.56	13.20	13.28	13.17	13.67	13.44	12.82	12.85	13.13	15.02	16.22	16.93	15.33	16.28	15.07	15.21
FeO*	17.68	18.44	18.82	19.39	20.17	13.24	20.70	22.61	21.42	19.08	20.41	17.94	26.70	28.22	28.54	27.50
MnO	0.43	0.38	0.21	0.15	0.14	0.09	0.20	0.25	0.19	0.15	0.62	0.43	0.70	0.84	0.99	0.88
MgO	14.83	13.86	13.45	12.66	12.31	16.83	12.26	11.05	11.59	13.29	11.27	11.63	5.52	4.56	5.13	4.99
Na_2O	0.09	0.05	0.13	0.13	0.09	0.22	0.13	0.09	0.21	0.21	0.10	0.14	0.04	0.13	0.15	0.12
K_2O	9.60	9.55	9.78	9.32	9.57	9.89	9.44	9.41	9.51	9.15	9.70	9.59	9.31	9.35	9.12	9.14
Total	98.89	97.82	99.10	96.92	98.64	98.90	98.07	97.76	99.02	97.80	98.81	97.21	97.48	98.67	98.68	96.84
Formula (apfu)																
Si	5.569	5.645	5.555	5.514	5.508	5.575	5.580	5.588	5.609	5.376	5.440	5.446	5.578	5.511	5.538	5.550
Ti	0.503	0.472	0.615	0.622	0.621	0.670	0.588	0.530	0.578	0.532	0.422	0.436	0.449	0.408	0.445	0.427
Al (iv)	2.431	2.355	2.445	2.486	2.492	2.425	2.420	2.412	2.391	2.624	2.560	2.554	2.422	2.489	2.462	2.450
Al (vi)	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.026	0.023	0.273	0.411	0.374	0.465	0.282	0.362
Fe as Fe^{2+}	2.160	2.286	2.311	2.444	2.502	1.583	2.591	2.868	2.661	2.376	2.530	2.230	3.458	3.635	3.687	3.610
Mn	0.053	0.048	0.026	0.020	0.017	0.011	0.025	0.032	0.024	0.019	0.078	0.054	0.092	0.110	0.130	0.116
Mg	3.230	3.063	2.941	2.842	2.722	3.587	2.735	2.498	2.565	2.942	2.489	2.576	1.274	1.047	1.181	1.167
Na	0.026	0.016	0.037	0.037	0.026	0.061	0.038	0.026	0.061	0.060	0.029	0.041	0.012	0.038	0.045	0.038
K	1.789	1.806	1.832	1.792	1.811	1.804	1.802	1.821	1.801	1.736	1.833	1.818	1.838	1.837	1.797	1.829
O	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22

	Phnom Baseth South		Dar		Phnom Aural North		Phnom Aural South			Kirivong		Phnom Tamao		Phnom Da		Phnom Rat Bat	Svay Leu
Sample No.	CG305	CG306	CG307	CG309	CG311	CG314	CG315	CG316	CG317	CG319	CG320	CG322	CG323	CG406	CG408	CG410	CG412
Number of samples	5	5	5	5	5	3	5	5	5	5	5	5	5	1	4	1	5
Oxide(wt%)																	
SiO_2	36.26	36.04	35.08	34.75	34.43	35.31	35.44	35.19	35.36	35.76	36.52	35.80	35.92	33.90	37.89	34.31	38.99
TiO_2	3.84	3.94	4.37	4.49	3.29	3.05	3.68	4.09	3.97	2.68	2.10	3.50	3.64	1.77	2.13	4.84	3.55
Al_2O_3	14.08	14.15	12.84	13.36	13.72	13.98	13.90	13.72	13.47	16.06	18.37	15.05	15.67	14.99	17.93	12.69	14.39
FeO*	27.70	27.83	32.09	28.30	31.69	31.97	30.84	28.70	30.80	30.42	28.60	29.60	28.58	34.63	28.44	33.11	15.70
MnO	0.80	0.59	1.40	1.19	0.84	0.96	0.54	0.42	0.49	0.38	0.44	0.62	0.49	1.06	1.03	0.47	0.54
MgO	5.85	5.77	4.42	6.47	4.94	4.22	5.62	6.30	5.29	3.15	2.62	4.48	4.61	1.45	1.99	3.96	15.80
Na_2O	0.08	0.05	0.16	0.14	0.09	0.08	0.04	0.08	0.06	0.17	0.27	0.09	0.08	0.05	0.15	0.21	0.19
K_2O	9.10	9.15	8.56	9.20	8.88	8.96	9.18	9.10	9.02	9.39	9.78	9.20	9.34	8.11	8.65	8.63	9.04
Total	97.71	97.53	98.92	97.91	97.88	98.52	99.23	97.59	98.46	98.00	98.70	98.34	98.34	95.96	98.22	98.23	98.21
Formula (apfu)																	
Si	5.632	5.612	5.524	5.453	5.474	5.568	5.509	5.514	5.542	5.597	5.603	5.570	5.555	5.558	5.790	5.470	5.647
Ti	0.448	0.461	0.518	0.530	0.393	0.362	0.430	0.482	0.468	0.315	0.244	0.410	0.424	0.218	0.245	0.580	0.387
Al (iv)	2.368	2.388	2.476	2.547	2.526	2.432	2.491	2.486	2.458	2.403	2.397	2.430	2.445	2.442	2.210	2.530	2.353
Al (vi)	0.209	0.208	0.000	0.000	0.044	0.166	0.056	0.048	0.057	0.559	0.925	0.330	0.411	0.456	1.019	0.000	0.102
Fe as Fe^{2+}	3.601	3.625	4.227	3.714	4.213	4.216	4.010	3.760	4.037	3.982	3.670	3.853	3.696	4.749	3.634	4.415	1.909
Mn	0.105	0.078	0.187	0.158	0.113	0.128	0.071	0.056	0.065	0.050	0.057	0.081	0.065	0.147	0.134	0.064	0.067
Mg	1.355	1.339	1.038	1.514	1.171	0.991	1.303	1.471	1.234	0.736	0.598	1.040	1.063	0.354	0.453	0.942	3.409
Na	0.023	0.017	0.047	0.041	0.028	0.025	0.011	0.024	0.019	0.053	0.081	0.027	0.023	0.015	0.044	0.064	0.053
K	1.804	1.817	1.721	1.842	1.800	1.802	1.821	1.819	1.804	1.876	1.913	1.825	1.843	1.697	1.687	1.755	1.672
O	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22	22

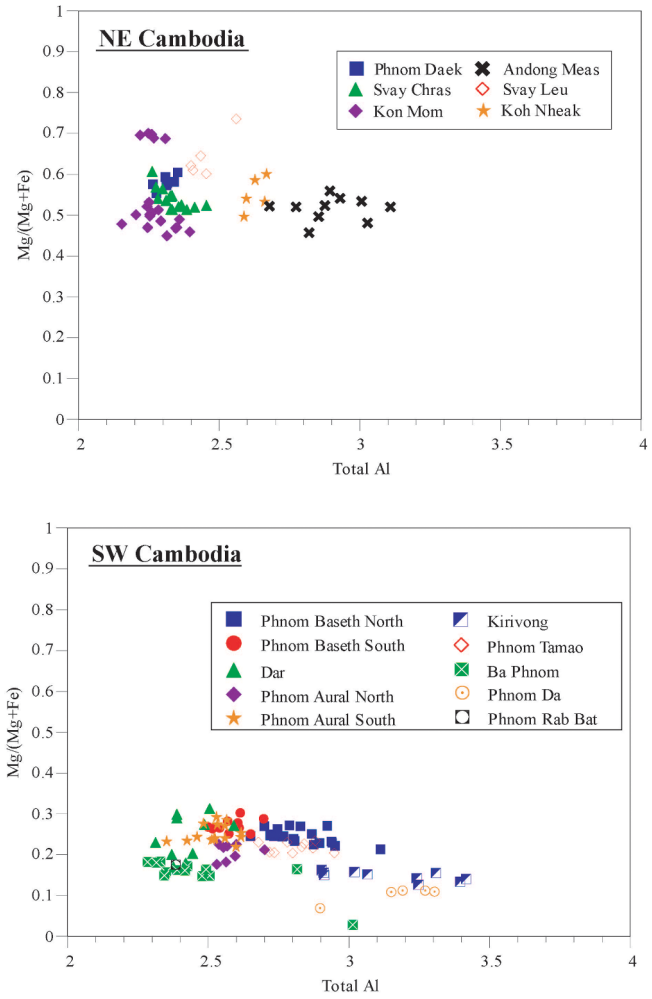


Fig. 14. Relationship between Mg/(Mg + Fe) molar ratio and total Al content in biotite for the igneous rocks from Cambodia.

Dinhquan granitic rocks consist of hornblende-biotite granodiorite, diorite, and minor granite emplaced at 112–100 Ma. They suggest that the granitic rocks in the Dalat-Kratie zone were formed by the subduction of the Paleo-Pacific Ocean crust beneath Indochina during the Cretaceous.

The igneous rocks in the SW region from the Late Triassic to the Cretaceous (the Chanthaburi-Kampong Chhnang zone) are of the ilmenite series; the rock samples from Phnom Aural North and South, Dar, Phnom Rab Bat and Chumkiri are I-type; those from Phnom Baseth North and South are I-S type; and the Phnom Tamao and Ba Phnom samples are A-type. These igneous rocks have SiO_2 vs. ($Na_2O + K_2O$) contents, REE patterns, non-adakitic signatures, and tectonic setting discrimination similar to those of the granitic rocks in the Loei fold belt located south of the Mae Ping fault in Thailand that were emplaced in the Late Triassic to Early Jurassic (207–195 Ma) (Charusiri et al., 1993; Zaw et al., 2009; Kawakami et al., 2014; Nualkhao et al., 2018). They were formed by the subduction of the Paleo-Tethys Ocean crust associated with the collision of the Sibumasu Terrane with the Indochina Terrane.

The A-type granites in Kirivong and Phnom Da with formation ages of 71 ± 6 Ma and 58 ± 4 Ma, respectively, may correspond to the Cana granitic rocks in the Dalat-Kratie zone in Vietnam. This zone has biotite-bearing granites with low hornblende content, a strong negative Eu anomaly, and ages around 96–93 Ma (Thuy et al., 2004). Shellnutt et al. (2013) concluded that the Cretaceous Cana granitic rocks (the Ankroet granitic rocks) are post-orogenic/within-plate granites (A-type), and that they intruded in association with the subduction of the Paleo-Pacific Ocean crust beneath Indochina during the Cretaceous to the Paleogene.

5. Conclusions

Igneous rocks in Cambodia can be divided by the inferred Mae Ping fault into two regions: the NE region and the SW region.

All the igneous rocks in the NE region are of the magnetite series and can be subdivided into adakitic rocks and non-adakitic rocks. The adakitic rocks are of I-type and formed in a volcanic arc setting. They are almost metaluminous. No negative Eu anomaly suggests that the magma did not undergo plagioclase fractionation or experience relatively reduced conditions. The magma sources of the adakitic rocks were derived from the mantle with compositions near PM. The older adakitic rocks aged 332 ± 5 Ma (Koh Nheak, Andong Meas, Oyadav and Svay Chras (CG208)) were formed by the subduction of the young Paleo-Tethys Ocean crust between the Kontum block and the Khorat block beneath the latter in the Middle Carboniferous (the Early Indosinian Orogeny). The younger adakitic rocks with ages ranging from 274 Ma to 268 Ma (Phnom Daek, Phnom Koy Rmeas, and Svay Leu) have the same geochemical signatures and magma source as the older adakitic rocks. The younger adakitic igneous rocks may have been emplaced by the subduction of the young branch of the Paleo-Tethys Ocean crust between the Khorat block and the Cambodia-offshore Malaysia block beneath the latter (the Indosinian Orogeny I). These older and younger adakitic rocks are associated with Cu, Fe, and/or Au mineralization. The syenite-diorite from Kon Mom and Svay Chras, and granodiorite from Phnom Soporkaley, are non-adakitic, of magnetite series and volcanic to *syn*-collision setting. The former two are classified as A-type and the latter as I-type. Their formation ages are 103 ± 2 Ma, 97 ± 14 Ma, and 118 ± 30 Ma, respectively, and they show a slightly negative Eu anomaly. These rocks were formed by the subduction of the Paleo-Pacific Ocean crust beneath Indochina during the Cretaceous.

The igneous rocks in the SW region are of the ilmenite series, non-adakitic, and metaluminous to slightly peraluminous. They were derived from lower to middle continental crust materials with a relatively uniform Nd isotope ratio. They have higher contents of incompatible elements such as Rb, Y, Sn, U, Pb, and Th than the igneous rocks in the NE region. They are subdivided into I-type, I-S-A-type, and A-type with two different ages and tectonic settings. The Late Triassic to the Middle Jurassic (205–167 Ma) granites in Dar, Phnom Rab Bat, Phnom Aural North and South, Ratanakmundol, and Chumkiri are of I-type, and

Table 4
Results of Sr and Nd isotopic analyses for igneous rock samples collected in this study.

Location	Sample No.	$^{87}Sr/^{86}Sr$	Std. Error	$^{87}Rb/^{86}Sr$	Age (Ma)	MSWD	$(^{87}Sr/^{86}Sr)_i$	$^{143}Nd/^{144}Nd$	Std. Error	$^{147}Sm/^{144}Nd$	$(^{143}Nd/^{144}Nd)_i$
Phnom Daek	CG201	0.705020	0.000006	0.327963	273.9 ± 15.4		0.703811	0.512784	0.000004	0.190195	0.512462
	CG202	0.704741	0.000030	0.326149		18	0.703539	0.512856	0.000003	0.135938	0.512626
Phnom Koy Rmeas	CG204	0.703727	0.000064	0.012594			0.703680	0.512957	0.000004	0.176536	0.512658
Kon Mom	CG210	0.704926	0.000011	0.879318	103.37 ± 2.47		0.703614	0.512825	0.000004	0.125272	0.512739
	CG211	0.705530	0.000007	1.316429			0.703565	0.512779	0.000003	0.114037	0.512700
	CG212	0.705023	0.000007	0.948033		2.1	0.703609	0.512811	0.000003	0.135402	0.512718
	CG213	0.705851	0.000007	1.502893			0.703609	0.512785	0.000004	0.120910	0.512702
	CG214	0.704896	0.000008	0.889561			0.703568	0.512790	0.000004	0.129992	0.512701
Svay Chras	CG206	0.704443	0.000008	0.333516	97.3 ± 14.2		0.704031	0.512787	0.000003	0.129547	0.512714
	CG207A	0.704288	0.000010	0.233892		10	0.703999	0.512791	0.000003	0.129602	0.512717
	CG207B	0.704385	0.000008	0.336755			0.703969	0.512789	0.000002	0.134479	0.512713
	CG208	0.704516	0.000008	0.180696	331.9 ± 4.77	28	0.703690	0.512754	0.000003	0.122327	0.512497
Koh Nhaek	CG215	0.704118	0.000012	0.093708			0.703690	0.512897	0.000004	0.186621	0.512505
	CG216	0.704134	0.000008	0.103724			0.703660	0.512908	0.000004	0.193994	0.512501
	CG217	0.703744	0.000006	0.070004			0.703424	0.512899	0.000004	0.175806	0.512529
Andong Meas	CG218	0.704594	0.000006	0.179093			0.703776	0.512707	0.000003	0.118409	0.512458
	CG219	0.704540	0.000007	0.223414			0.703520	0.512691	0.000003	0.109363	0.512461
Oyadav	CG220	0.706781	0.000009	0.665801			0.703740	0.512627	0.000006	0.121339	0.512372
	CG221	0.706693	0.000006	0.684835			0.703564	0.512644	0.000004	0.120576	0.512390
	CG222	0.706671	0.000007	0.688879			0.703524	0.512637	0.000003	0.124062	0.512377
Svay Leu	CG412	0.704669	0.000012	0.338882	267.6 ± 33.2	N/A	0.703368	0.512899	0.000006	0.123197	0.512681
	CG413	0.703927	0.000013	0.145776			0.703367	0.512926	0.000004	0.144485	0.512671
Phnom Soporkaley	CG414	0.707216	0.000011	1.251637	117.9 ± 29.7	N/A	0.705099	0.512792	0.000004	0.144201	0.512680
	CG415	0.706978	0.000014	1.110868			0.705100	0.512797	0.000004	0.146774	0.512683
Phnom Basset North	CG301	0.765193	0.000009	23.089431	167.06 ± 8.47		0.706759	0.512615	0.000009	0.145485	0.512446
	CG302	0.772429	0.000007	26.450903			0.705487	0.512656	0.000010	0.150823	0.512480
	CG303	0.776038	0.000009	27.203053		1.2	0.707192	0.512650	0.000008	0.160700	0.512463
Phnom Basset South	CG305	0.733206	0.000008	10.198028			0.707397	0.512640	0.000007	0.174048	0.512437
	CG306	0.745048	0.000009	16.078185			0.704357	0.512597	0.000009	0.133119	0.512442
Dar	CG307	0.745730	0.000009	14.374382	189.9 ± 15.2	0.61	0.705054	0.512574	0.000004	0.115823	0.512423
	CG309	0.722989	0.000005	6.253413			0.705293	0.512621	0.000007	0.119645	0.512466
Phnom Rab Bat	CG409	0.727403	0.000012	7.867242			0.705140	0.512555	0.000003	0.126563	0.512390
	CG410	0.727136	0.000012	8.065880			0.704311	0.512543	0.000003	0.124998	0.512380
Phnom Aural North	CG311	0.732274	0.000008	9.785585	197.53 ± 4.80		0.706812	0.512568	0.000007	0.137439	0.512403
	CG314	0.772946	0.000011	25.231994			0.707293	0.512633	0.000009	0.126979	0.512481
Phnom Aural South	CG315	0.716369	0.000009	3.509508		4.8	0.707237	0.512457	0.000007	0.122959	0.512310
	CG316	0.711520	0.000007	1.831423			0.706755	0.512474	0.000007	0.123456	0.512326
	CG317	0.712836	0.000008	2.049354			0.707503	0.512444	0.000006	0.129375	0.512289
Chumkiri	CG318	0.718607	0.000006	3.669383			0.709060	0.512524	0.000007	0.117974	0.512383
Kirivong	CG319	0.717981	0.000007	10.572328	70.56 ± 5.58	N/A	0.707316	0.512630	0.000005	0.143858	0.512563
	CG320	0.727357	0.000008	19.840541			0.707343	0.512629	0.000006	0.161453	0.512554
Phnom Tamao	CG322	0.770498	0.000006	22.950873	182.2 ± 43.3	N/A	0.710453	0.512569	0.000006	0.159827	0.512377
	CG323	0.773244	0.000010	24.001186			0.710451	0.512522	0.000008	0.133809	0.512360
Ba Phnom	CG401	0.736850	0.000007	9.630173	205.19 ± 4.88		0.710147	0.512490	0.000004	0.132531	0.512321
	CG402	0.752968	0.000008	14.654476			0.712334	0.512496	0.000003	0.133923	0.512325
	CG403	0.732915	0.000007	8.021460		1.7	0.710673	0.512493	0.000003	0.153042	0.512297
	CG404	1.082376	0.000013	136.940505			0.702663	0.512589	0.000003	0.203644	0.512329
	CG405	0.971903	0.000010	86.860036			0.731055	0.512540	0.000004	0.164458	0.512331
Phnom Da	CG406	0.814652	0.000010	109.547875	58.19 ± 3.56		0.722835	0.512621	0.000004	0.219664	0.512536
	CG407	1.141945	0.000272	503.930013		0.01	0.719575	0.512657	0.000005	0.306416	0.512539
	CG408	0.786073	0.000015	77.272223			0.721307	0.512560	0.000004	0.179760	0.512491

those in Phnom Baseth North and South (porphyritic) are of I-S-A-type or A-type. The I-type and I-S-A-type granites show a moderate negative Eu anomaly, with the A-type granites (Ba Phnom and Phnom Tamao) showing a strongly negative Eu anomaly. The igneous rocks in the SW region (the Chanthaburi–Kampong Chhnang zone) are similar to those emplaced in the Loei fold belt located south of the Mae Ping fault in Thailand of the Late Triassic to Early Jurassic (207–195 Ma), and were formed by the Paleo-Tethys Ocean subduction beneath the Indochina Terrane and the collision of the Sibumasu Terrane with the Indochina Terrane (the Indosinian Orogeny II). The younger A-type granites in Phnom Da and Kirivong have ages of 58 ± 4 Ma and 71 ± 6 Ma, respectively. These granites are enriched in REEs, and show a strongly negative Eu anomaly. They are post-orogenic, within-plate or rifting zone granites that intruded after the subduction of the Paleo-Pacific

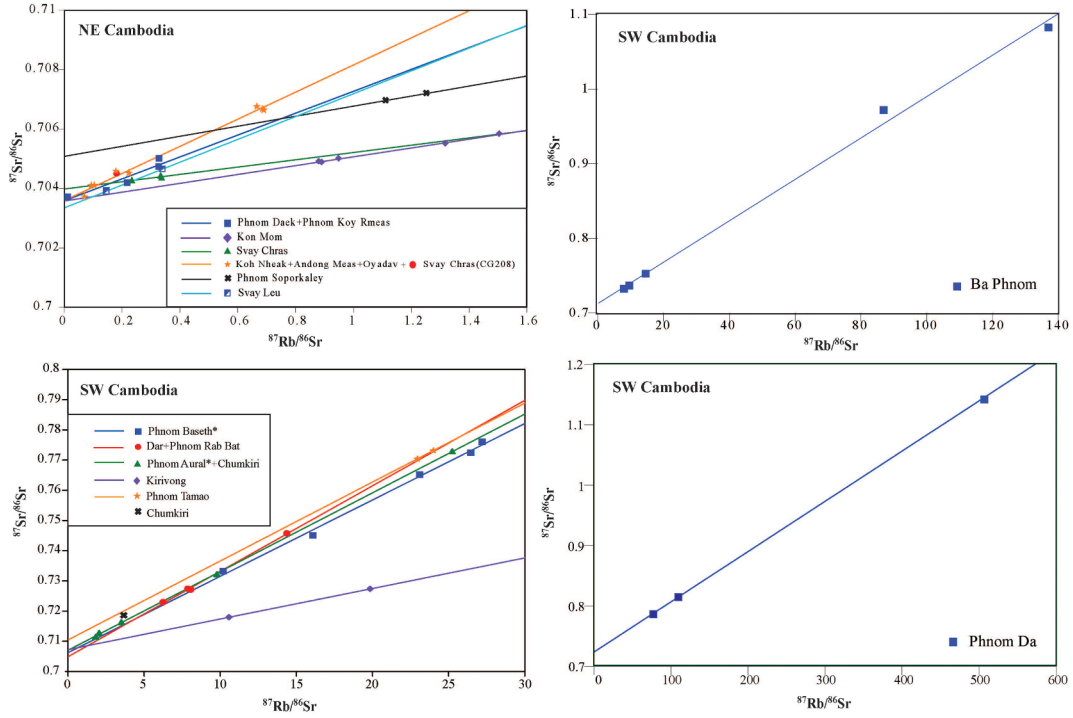


Fig. 15. Rb-Sr isochron diagram for the igneous rocks from Cambodia.

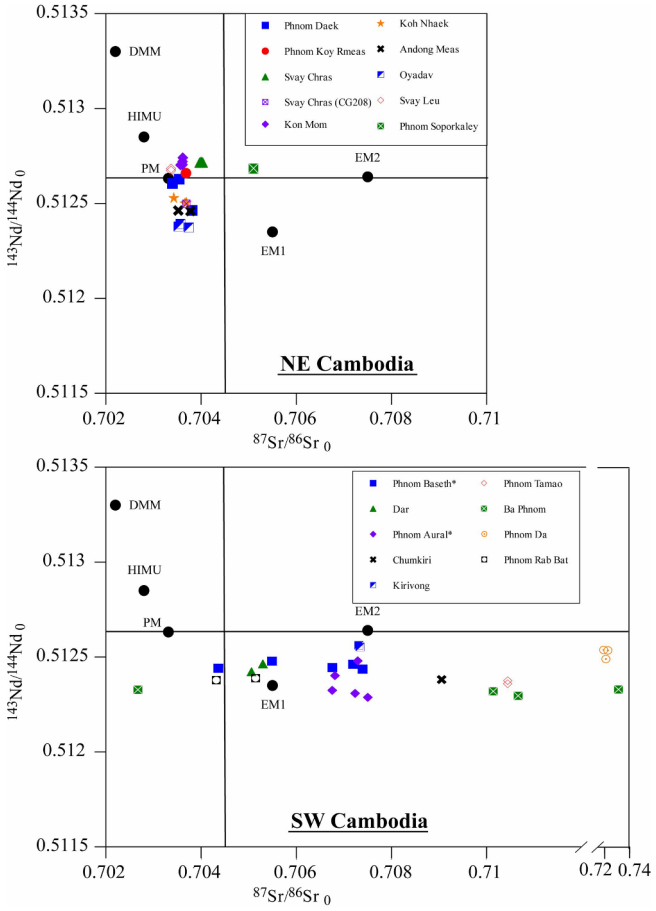


Fig. 16. $(^{87}Sr/^{86}Sr)_i$ vs. $(^{143}Nd/^{144}Nd)_i$ diagram for the igneous rocks from Cambodia.

Ocean crust beneath Indochina during the Cretaceous to the Paleogene.

6. Author agreement

Uchida and Kong designed this study. Cheng, Katayose, Yarimizu, Uchida and Nakano conducted the field survey. Cheng, Katayose, Yarimizu and Shin carried out the laboratory works. All the authors interpreted data. Cheng, Uchida and Nakano drafted the manuscript. All authors read and approved the final manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The first author, R. Cheng, would like to express his sincere gratitude to the Japan International Cooperation Agency (JICA) for the master's scholarship opportunity and research grant and to the Ministry of Mines and Energy of Cambodia that permitted and encouraged him to continue the research on his master's degree. We would like to express our thanks to Angkor Gold Corp who provided the drilled core samples from Koh Nheak, Andong Meas, and Oyadav, and to Rithy Granite Co., Ltd. for providing access to their quarries at Svay Chras. This study was conducted with the support of a Joint Research Grant for the Environmental Isotope Study of the Research Institute for Humanity and Nature, and also was financially supported in part by Grants-in-Aid for Scientific Research of the Japan Society for the Promotion of Science (Grant no. 16K06931 (E. Uchida)). The authors are grateful to journal reviewers, Dr. Takayuki Manaka, Dr. Abhisit Salam and an anonymous reviewer for their insightful reviews and valuable comments to improve the quality of the manuscript as well as Prof Khin Zaw for editorial input and suggestion. We thank Edanz Editing China for editing the English text of a draft of this manuscript.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jseaes.2019.104046.

References

Bonin, B., 2007. A-type granites and related rocks: Evolution of a concept, problems and prospects. Lithos 97, 1–29.

Carter, A., Roques, D., Bristow, C., Kinny, P., 2001. Understanding Mesozoic accretion in Southeast Asia: significance of Triassic thermotectonism (Indosinian orogeny) in Vietnam. Geology 29, 211–214.

Chappell, B.W., White, A.J.R., 1974. Two contrasting granite types. Pac. Geol. 8, 173–174.

Chappell, B.W., White, A.J.R., 2001. Two contrasting granite types: 25 years later. Aust. J. Earth Sci. 48, 489–500.

Charusiri, P., Clark, A.H., Farrar, E., Archibald, D., Charusiri, B., 1993. Granite belts in Thailand: evidence from $^{40}Ar/^{39}Ar$ geochronological and geological syntheses. J. Southeast Asian Earth Sci. 8, 127–136.

Cox, K.G., Bell, J.D., Pankhurst, R.J., 1979. The Interpretation of Igneous Rocks. Allen and Unwin, London, pp. 450.

Defant, M.J., Drummond, M.S., 1990. A model for Trondhje-Tonalite-Dacite Genesis and crustal growth via slab melting: Archean to modern comparisons. J. Geophys. Res.: Solid Earth Banner 95, 21503–21521.

Dickin, A.P., 2005. Radiogenic Isotope Geology. Cambridge University Press, pp. 508.

ESCAP, 1993. Atlas of Mineral Resources of the ESCAP region: Geological map of Cambodia; Mineral resources map of Cambodia. vol.10. United Nations, New York.

Faure, M., Nguyen, V.V., Hoai, L.T.T., Lepvrier, C., 2018. Early Paleozoic or Early-Middle Triassic collision between the South China and Indochina Blocks: The Controversy resolved? Structural insights from the Kon Tum Massif (Central Vietnam). J. Asian Earth Sci. 166, 162–180.

Fontaine, H., Workman, D., 1978. Review of the geology and mineral resources of Kampuchea, Laos and Vietnam. In: Proceeding of 3rd Conference of Geology and Mineral Resources in Southeast Asia, Bangkok, pp. 541–603.

Fyhn, M.B.W., Green, P.F., Bergman, S.C., Van Itterbeeck, J., Tri, T.V., Dien, P.T., Abatzis, I., Thomsen, T.B., Chea, S., Pedersen, S.A.S., Mai, L.C., Tuan, H.A., Nielsen, L.H., 2016. Cenozoic deformation and exhumation of the Kampot Fold Belt and implications for south Indochina tectonics. J. Geophys. Res. Solid Earth 121, 5278–5307.

Gill, R., 2010. Igneous Rocks and Processes: A Practical Guide. Wiley-Blackwell, pp. 438.

Ishihara, S., 1977. The magnetite-series and ilmenite-series granitic rocks. Min. Geol. 27, 293–305.

Kamvong, T., Zaw, Khin, Meffre, S., Maas, R., Stein, H., Lai, C.K., 2014. Adakites in the Truong Son and Loei fold belts, Thailand and Laos: Genesis and implications for geodynamics and metallogeny. Gondwana Res. 26, 165–184.

Kawakami, T., Nakano, N., Higashino, F., Hokada, T., Osanai, Y., Yuhara, M., Charusiri, P., Kamikubo, H., Yonemura, K., Hirata, T., 2014. U-Pb zircon and CHIME monazite dating of granitoids and high-grade metamorphic rocks from the Eastern and Peninsula Thailand – A new report of early Paleozoic granite. Lithos 200, 64–79.

Kong, S., Watanabe, K., Imai, A., Gabbo, J., Scherstén, A., 2011. Magnetic Susceptibility, Geology, Petrography, and Geochemistry of the Granite Host Rocks of Baseth Mo-Cu mineralized stock, Cambodia: implication for its origin. In: Proceedings of GSA Annual Meeting in Minneapolis.

Kong, S., Watanabe, K., Imai, A., 2012. Magnetic susceptibility and geochemistry of granitic rocks in Cambodia. ASEAN Eng. J., Part C 2, 113–132.

Lepvrier, C., Maluski, H., Tich, V.V., Leyreloup, A., Thi, P.T., Vuong, N.V., 2004. The Early Triassic Indosinian orogeny in Vietnam (Truong Son Belt and Kontum Massif); implications for the geodynamic evolution of Indochina. Tectonophysics 393, 87–118.

Lepvrier, C., Vuong, N.V., Maluski, H., Thi, P.T., Vu, T.V., 2008. Indosinian tectonics in Vietnam. Comptes Rendus – Geosci. 340, 94–111.

Loiselle, M.C., Wones, D.R., 1979. Characterization and origin of anorogenic granites. Geol. Soc. Am., Abstr. Progr 11, 468.

McDonough, W.F., Sun, S.S., 1995. The composition of the Earth. Chem. Geol. 120, 223–253.

Metcalfe, I., 2013. Gondwana dispersion and Asian accretion: Tectonic and palaeogeographic evolution of eastern Tethys. J. Asian Earth Sci. 66, 1–33.

Metcalfe, I., 2011. Palaeozoic–Mesozoic history of SE Asia. Geological Society, London, Special Publications 355, pp. 7–35.

Morley, C.K., 2012. Earth-science reviews late cretaceous – early palaeogene tectonic development of SE Asia. Earth-Sci. Rev. 115, 37–75.

Morley, C.K., Ampaiwan, P., Thanudamrong, S., Kuenphan, N., Warren, J., 2013. Development of the Khao Khwang Fold and Thrust Belt: Implications for the geodynamic setting of Thailand and Cambodia during the Indosinian Orogeny. J. Asian Earth Sci. 62, 705–719.

Nédélec, A., Bouchez, J.L., 2015. Granites – Petrology, Structure, Geological Setting, and Metallogeny. Oxford University Press, United Kingdom, pp. 334.

Nualkhao, P., Takahashi, R., Imai, A., Charusiri, P., 2018. Petrochemistry of granitoids along the Loei Fold Belt, Northeastern Thailand. Resour. Geol. 68, 395–424.

Pearce, J.A., Harris, N.B.W., Tindle, A.G., 1984. Trace element distribution diagrams for the tectonic interpretation of granitic rocks. J. Petrol. 25, 956–983.

Rossignol, C., Bourquin, S., Hallot, E., Poujol, M., Dabard, M.P., Martini, R., Villeneuve, M., Cornée, J.J., Brayard, A., Roger, F., 2018. The Indosinian orogeny: a perspective from sedimentary archives of north Vietnam. J. Asian Earth Sci. 158, 352–380.

Sanematsu, K., Murakami, H., Duangsurigna, S., Vilayhack, S., Duncan, R.A., Watanabe, Y., 2011. $^{40}Ar/^{39}Ar$ ages of granitoids from the Truong Son fold belt and Kontum massif in Laos. J. Mineral. Petrol. Sci. 106, 13–25.

Schaefer, B.F., 2016. Radiogenic Isotope Geochemistry. Oxford University Press, United Kingdom, pp. 196.

Shellnutt, J.G., Lan, C.Y., Long, T.V., Usuki, T., Yang, H.J., Mertzman, S.A., Iizuka, Y., Chung, S.L., Wang, K.L., Hsu, W.Y., 2013. Formation of Cretaceous Cordilleran and post-orogenic granites and their microgranular enclaves from the Dalat zone, southern Vietnam: Tectonic implications for the evolution of Southeast Asia. Lithos 182–183, 229–241.

Shin, K.C., Kurosawa, M., Anma, R., Nakano, T., 2009. Genesis and mixing/mingling of mafic and felsic magmas of back arc granite: Miocene Tsushima pluton, Southeast Japan. Resour. Geol. 59, 25–50.

Sone, M., Metcalfe, I., 2003. Parallel Tethyan Sutures in mainland SE Asia: new insights for Palaeo-Tethys closure. Compte Rendus Geosci. 340, 166–179.

Thuy, N.T.B., Satir, M., Siebel, W., Vennemann, T., Long, T. Van, 2004. Geochemical and isotopic constraints on the petrogenesis of granitoids from the Dalat zone, southern Vietnam. J. Asian Earth Sci. 23, 467–482.

Tien, P.C., An, L.D., Bach, L.D., Bac, D.D., Vongdara, B., Phengthavongsa, B., Danh, T., Dy, N.D., Dung, H.T., Hai, T.Q., Khuc, V., Kun, S.C., Long, P.D., Ly, M.N., My, N.Q., Ngan, P.K., Ngoc, N., Ratanavong, N., Quoc, N.K., Quyen, N.V., Aphaymani, S.D., Thanh, T.D., Tri, T.V., Truyen, M.T., Xay, T.S., 1990. Geology of Cambodia, Laos and Vietnam. Explanatory note to the geological map of Cambodia, Laos and Vietnam at 1: 1,000,000 scale. The Geological Survey of Vietnam, Hanoi.

Vermeesch, P., 2019. IsoplotR: a free and open toolbox for geochronology. Geosci. Front. 9, 1479–1493.

Whalen, J.B., Kenneth, C.L., Chappell, B.W., 1987. A-type granites: geochemical characteristics, discrimination and petrogenesis. Contrib. Miner. Petrol. 95, 407–419.

Wilson, M., 1989. Igneous Petrogenesis. In: A Global Tectonic Approach. Unwin Hyman, London, pp. 466.

Workman, D.R., 1977. Geology of Laos, Cambodia, South Vietnam and the eastern part of Thailand. Overseas Geol. Min. Resour. 50, 33pp.

Zaw, Khin, Meffre, S., Kamvong, T., Stein, H., Vasconcelos, P., Golding, S., 2009. Geochronological and metallogenic framework of Cu-Au skarn deposits along Loei Fold Belt, Thailand and Lao PDR. In: Proceeding of the Tenth Biennial SGA Meeting, Townsville, pp. 309–311.

Zaw, Khin, Meffre, S., Lai, C.K., Burrett, C., Santosh, M., Graham, I., Manaka, T., Salam, A., Kamvong, T., Cromie, P., 2014. Tectonics and metallogeny of mainland Southeast Asia – A review and contribution. Gondowana Res. 26, 5–30.